CLAIM With gaps D then Δ along the x-axis,

$$S_{\text{first}} = \frac{|a|D^3}{12}(D + 2\Delta), \qquad S_{\text{second}} = \frac{|a|\Delta^3}{12}(\Delta + 2D).$$

When the three intersections are equally spaced, both collapse to $|a|d^4/4$.

WHY You do not need either formula if you own the splitting trick. Write the far factor in terms of the near one: $x - \gamma = (x - \beta) - \Delta$. That turns the integral into $\int (x-\alpha)(x-\beta)^2 - \Delta \int (x-\alpha)(x-\beta)$, and both pieces are in the table above: $D^4/12 + \Delta D^3/6$.

COST About twenty seconds either way. The trick is worth more than the formula because it also handles the cases nobody tabulated.

BREAKS **The lobes are not equal unless the gaps are.** Assuming symmetry because the picture looks symmetric is the single most common error on this configuration.

DEFAULT Integrate the expanded cubic across each interval separately.

54.3 Chord and tangents: the 2 to 1 ratio

54.3 THE PARABOLA SPLITS ITS CHORD REGION 2:1 EDGE

TRIGGER A parabola with tangents drawn at two points, meeting somewhere below or above the chord.

CLAIM The region between the parabola and the chord is exactly **twice** the region between the parabola and the two tangents. The tangents meet at $x_R = \frac{1}{2}(x_P + x_Q)$, and the tangent at x_R is parallel to the chord PQ.

WHY The gradient of a quadratic is linear in x, so the gradients at x_P, x_R, x_Q are in arithmetic progression and the middle one is the mean, which is the chord's gradient. The areas then follow from two applications of $|a|D^3/6$ on half-width intervals.

COST Turns a two-tangent area question into one subtraction. Perhaps one question in three papers.

BREAKS Parabolas only. There is no 2 to 1 law for a cubic.

DEFAULT Find both tangents, find the intersection, integrate.

DRILL A **RECOGNITION ONLY. NAME THE CONSTANT AND THE ANSWER, TEN SECONDS EACH.**

1. $y = 3x^2 + \ldots$ meets a line at $x = 2$ and $x = 4$.
2. $y = -\frac{1}{6}x^2 + \ldots$ meets $y = -2$ at $x = -981$ and $x = -975$.
3. $y = 8(x - 126)^2(x - 129)$, area to $y = 0$.
4. $y = 2^{-33}x^2 + \ldots$ cut by a line at $x = \pm 2^{10}$.
5. $y = a(x - \alpha)^2(x - \beta)^2$, area to the line through both touch points.

ANSWERS. 1. $\frac{1}{6} \cdot 3 \cdot 2^3 = 4$. 2. $\frac{1}{6} \cdot \frac{1}{6} \cdot 6^3 = 6$. 3. $\frac{1}{12} \cdot 8 \cdot 3^4 = 54$. 4. $\frac{2^{-33}}{6}(2^{11})^3 = \frac{2^{-33} \cdot 2^{33}}{6} = \frac{1}{6}$. 5. $|a|D^5/30$.

DRILL B **EXAM STANDARD, NINETY SECONDS EACH.**

1. The line $y = 2x + 1$ meets $y = x^3 + px + q$ at $x = a$, $x = a + 2$ and $x = a + 5$. Find the total area enclosed between the line and the curve.
2. A line is tangent to $y = -2x^3 + \ldots$ at a local maximum and cuts it again 3 units away in x. Find the enclosed area.

3. The area between $y = kx^2 + \dots$ and a line through its intersections at $x = t$ and $x = t + 6$ is 36. Find k.

ANSWERS. 1. Two lobes with $D = 2$ and $\Delta = 3$, leading coefficient 1: $\frac{1}{12} \cdot 2^3(2 + 2 \cdot 3) + \frac{1}{12} \cdot 3^3(3 + 2 \cdot 2) = \frac{8 \cdot 8}{12} + \frac{27 \cdot 7}{12} = \frac{16}{3} + \frac{63}{4} = \frac{253}{12}$. 2. Tangent to a cubic and one simple cut: $\frac{1}{12} \cdot 2 \cdot 3^4 = \frac{81}{6} = \frac{27}{2}$. 3. $|k| \cdot 216/6 = 36$ gives $|k| = 1$.

DRILL C THE 9.0 LAYER, THREE MINUTES.

1. The line $y = mx + c$ meets $y = x^3 - 3x^2 + 2x$ at three distinct points whose x-coordinates are in arithmetic progression. Find c in terms of m.
2. $y = x^3 - 3h^2x$ with $h > 0$. A horizontal line through the local maximum meets the curve again at $x = k$. Find k, and hence show that the five key points of a cubic are equally spaced in x.

ANSWERS. 1. The three intersections are the roots of $x^3 - 3x^2 + (2 - m)x - c = 0$, so their sum is 3 whatever the line does (§73). In an arithmetic progression the middle term is the mean, so the middle root is 1, and $f(1) = m + c$ gives $0 = m + c$, so $c = -m$. 2. Local maximum at $x = -h$, value $2h^3$; solving $x^3 - 3h^2x = 2h^3$ factorises as $(x + h)^2(x - 2h) = 0$, so $k = 2h$. The five points sit at $-2h, -h, 0, h, 2h$.

55 Curvature, the trapezium rule, and honest estimates

A small section that answers a question type appearing regularly and covered almost nowhere: *is this estimate too big or too small?*

55.1 Which way a curve bends

55.1 THE SECOND DERIVATIVE IS A SHAPE, NOT A NUMBER CORE

TRIGGER	"Is the estimate an overestimate or an underestimate", "sketch the curve", "explain why the chord lies above the curve", or a stationary point to classify.
CLAIM	Where $f'' > 0$ the curve is **convex** (bending upwards, holding water) and every chord lies **above** the curve. Where $f'' < 0$ it is **concave** and every chord lies **below**. A stationary point with $f'' > 0$ is a minimum and with $f'' < 0$ a maximum; if $f'' = 0$, classify by the sign of f' either side instead.
WHY	f'' is the rate of change of the gradient, so $f'' > 0$ means the gradient is increasing, which is exactly what "bending upwards" means.
COST	It answers the estimate question with no calculation, and it classifies stationary points faster than a sign table.
BREAKS	$f'' = 0$ does **not** imply a point of inflexion, and points of inflexion are not examined beyond the qualitative fact that a cubic has one. When $f'' = 0$ at a stationary point, fall back on the sign of f'.
DEFAULT	Evaluate the gradient just either side of the point.

55.2 The trapezium rule, and the only question it is ever asked in

55.2 CHORDS ABOVE, CHORDS BELOW EDGE

TRIGGER	A numerical estimate of an area, usually with the words "trapezium rule", followed by "overestimate or underestimate".

CLAIM	The trapezium rule replaces the curve by chords. So on an interval where the curve is **convex** ($f'' > 0$) the estimate is an **overestimate**; where the curve is **concave** ($f'' < 0$) it is an **underestimate**. More strips reduces the error but does not change the direction.
WHY	The trapezium is bounded above by the chord, and §55.1 tells you which side of the curve the chord is on.
COST	Ten seconds against computing the true integral to compare. This is essentially the only way this material is examined.
BREAKS	A curve that changes curvature inside the interval. Then the answer is "neither, in general", and the option list will contain exactly that.
DEFAULT	Compute both the estimate and the exact integral, when the exact one is available.

The same idea prices the other two standard estimates. A **left-endpoint rectangle rule** underestimates on an increasing function and overestimates on a decreasing one; the right-endpoint rule does the opposite. Neither depends on curvature, only on whether the function is going up or down, which is a useful contrast to hold beside the trapezium case.

55.3 Comparing two integrals without computing either

If $f(x) \geq g(x)$ on the whole of $[a, b]$, then $\int_a^b f \geq \int_a^b g$. That is obvious and it is the whole of a certain kind of question. The skill is establishing the pointwise inequality cheaply: compare the two functions at a few points, or subtract and complete the square, or notice that one is the other multiplied by something you can bound.

DRILL A RECOGNITION ONLY. OVER OR UNDER, TEN SECONDS EACH.

1. $f'' > 0$ across the interval. Is the trapezium estimate too big or too small?
2. $y = \ln x$ on $[1, 4]$. Over or under?
3. $f'(a) = 0$ and $f''(a) < 0$. What is the point?
4. $f'(a) = 0$ and $f''(a) = 0$. What do you do next?

ANSWERS. 1. Too big. Convex means every chord lies above the curve. 2. $f'' = -1/x^2 < 0$, concave, so an **under**estimate. 3. A maximum. 4. Classify by the sign of f' either side. $f'' = 0$ decides nothing on its own.

DRILL B EXAM STANDARD.

1. Is the trapezium rule estimate of $\int_1^2 x^2\,dx$ an over- or an underestimate?
2. Is the trapezium rule estimate of $\int_0^1 \sqrt{x}\,dx$ an over- or an underestimate?
3. Which is larger, $\int_0^1 x^2\,dx$ or $\int_0^1 x^3\,dx$, without computing either?
4. $f(x) = x^3 - 3x^2$. Classify both stationary points.

ANSWERS. 1. $f'' = 2 > 0$, convex, so an **overestimate**. 2. $f'' = -\frac{1}{4}x^{-3/2} < 0$ on $(0, 1]$, concave, so an **underestimate**. 3. On $(0, 1)$, $x^3 < x^2$, so $\int x^2$ is larger. 4. $f' = 3x^2 - 6x = 3x(x - 2)$, so $x = 0$ and $x = 2$; $f'' = 6x - 6$, giving $f''(0) = -6 < 0$, a maximum, and $f''(2) = 6 > 0$, a minimum.

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56 Calculus checkpoints

Four small results that between them cover most of the remaining calculus the paper leans on.

56.1 A sign change traps a root

56.1 CONTINUOUS, OPPOSITE SIGNS, THEREFORE A ROOT CORE

TRIGGER	"Show that the equation has a root between a and b", or "how many roots does this have", with a polynomial or another continuous function.
CLAIM	If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, there is at least one root strictly between them. Combine with monotonicity to get *exactly* one: if $f' > 0$ throughout, the root is unique.
WHY	A continuous graph cannot get from below the axis to above it without crossing.
COST	It is the intended route for the whole "show there is a root" family, and it takes two substitutions.
BREAKS	**At least one, not exactly one.** Two sign changes could hide three roots. And the converse fails: no sign change does not mean no root, because the curve can touch the axis and come back (§43.3).
DEFAULT	Solve, when you can, or sketch and count.

56.2 Where the maximum can be

On a closed interval the candidates for the greatest and least values are exactly: the stationary points **inside** the interval, and the **two endpoints**. Evaluate all of them, compare, done. Omitting the endpoints is a standard loss, and so is including a stationary point that lies outside the interval you were given.

56.3 Integrals as areas, with the sign

A definite integral is a **signed** area: the parts below the axis count negative. So "the area enclosed" and "the value of the integral" are different quantities whenever the curve crosses the axis inside the limits, and the paper knows that.

— For the **area**, split at every crossing and add the absolute values.

— For the **integral**, do not split; the signs are the answer.

— For the **area between two curves**, integrate the difference over each interval where the ordering is constant, and take the top curve minus the bottom one. The ordering can swap at an intersection, which is exactly where you split.

56.4 The integral that defines itself

56.4 AN UNKNOWN CONSTANT INSIDE AN INTEGRAL IS A SIMULTANEOUS EQUATION EDGE

TRIGGER	A definite integral whose integrand contains an unknown that is itself defined by an integral, or two integrals given as simultaneous conditions.
CLAIM	Give the unknown integral a name, say I, and integrate. Everything that does not depend on the variable of integration comes outside, and you are left with a linear equation in I. Solve for I.
WHY	A definite integral is a *number*, so it can be treated as a constant throughout the integration.
COST	Turns a question that looks circular into two lines. Uncommon, and unmistakable.

BREAKS Only when the "constant" actually depends on the variable of integration, which means you misread the limits. Check that the unknown is defined by a **definite** integral, with numbers at both ends.

DEFAULT There is not really one. Naming the constant is the question.

WORKED

"$f(x) = 3x^2 + k$ *where* $k = \displaystyle\int_0^1 f(x)\,dx$. *Find* k."

Integrate, treating k as the constant it is:

$$k = \int_0^1 \left(3x^2 + k\right) dx = \Big[x^3 + kx\Big]_0^1 = 1 + k.$$

That gives $k = 1 + k$, which is impossible, so **no such function exists** — and "no solutions" is a legitimate answer that option lists frequently include. Change the constant to $f(x) = 3x^2 + kx$ with $k = \int_0^1 f$ and you get $k = 1 + k/2$, so $k = 2$.

The point of showing both is that the method does not care which way it comes out. Name it, integrate, solve, and read what you got.

DRILL A **RECOGNITION ONLY. TEN SECONDS EACH.**

1. f is continuous with $f(1) = -2$ and $f(2) = 5$. What follows, and what does **not**?
2. Greatest value of f on $[0, 4]$. What is the complete list of candidates?
3. The curve crosses the axis inside the limits. Do "the area" and "the integral" agree?
4. $k = \displaystyle\int_0^2 (x + k)\,dx$. First move, and what does it come out as?

ANSWERS. 1. At least one root strictly between 1 and 2. Not that there is exactly one, and not that a sign change is necessary for a root. 2. The stationary points **inside** the interval, plus both endpoints. 3. No. The area splits at every crossing and takes absolute values. 4. Treat k as the constant it is: $k = \left[\frac{1}{2}x^2 + kx\right]_0^2 = 2 + 2k$, so $k = -2$.

DRILL B **EXAM STANDARD.**

1. Show that $x^3 + x - 3 = 0$ has a root between 1 and 2, and explain why it has exactly one real root.
2. Find the area enclosed between $y = x^3 - x$ and the x-axis.
3. Find the area between $y = x^2$ and $y = 2x$.
4. $f(x) = x^2 + c$ with $c = \int_0^2 f(x)\,dx$. Find c.

ANSWERS. 1. $f(1) = -1 < 0$ and $f(2) = 7 > 0$, and f is continuous, so there is a root between them; $f' = 3x^2 + 1 > 0$ everywhere, so f is strictly increasing and cannot cross twice. 2. Roots at $-1, 0, 1$; by symmetry (the function is odd) the two lobes have equal area, each $\int_0^1 (x - x^3)\,dx = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$, so the total **area** is $\frac{1}{2}$ — while the integral from -1 to 1 is 0. 3. They meet at $x = 0$ and $x = 2$, with the line above; $\int_0^2 (2x - x^2)\,dx = 4 - \frac{8}{3} = \frac{4}{3}$. Or use $|a|D^3/6 = 1 \cdot 8/6 = \frac{4}{3}$ (§54). 4. $c = \left[\frac{x^3}{3} + cx\right]_0^2 = \frac{8}{3} + 2c$, so $-c = \frac{8}{3}$ and $c = -\frac{8}{3}$.

12

PART XII

GEOMETRY

The specification lists the circle facts twice, once for coordinate geometry and once for plane geometry. What is examined is never the statement of a theorem. It is looking at a configuration and knowing, in two seconds, which fact it was built from.

57 Circles and configurations

58 Lines, distances and reflections

59 Triangle machinery, priced honestly

60 Awkward areas, and how to decompose them

Work the recognition drills against a clock. Everything here is a picture, and pictures are recognised or they are not.

57 Circles and configurations

57.1 The seven configurations

57.1 THE MOVE IS THE DRAWING, NOT THE THEOREM CORE

TRIGGER A circle diagram, or a circle described in words.

CLAIM Each of these is a trigger, and the move is a line you add. Adding the right line turns the question into angle chasing.

WHY All of them are consequences of one definition: every point of the circle is the same distance from the centre. Which is why the centre is where the information is, and why the first move is almost always to draw something to or through it.

COST The difference between thirty seconds and a stalled question. Recognition is the whole skill.

BREAKS Diagrams on screen are not to scale and cannot be marked up. What looks like a tangent may be a secant. **Take the wording as authoritative** and redraw on the board, see Book One §23.

DEFAULT Coordinates. Slower, but a circle question with numbers in it can always be done with $(x-a)^2+(y-b)^2=r^2$.

WHAT IS MENTIONED	DRAW THIS, AND READ OFF
A tangent	The radius to the point of contact. It is perpendicular.
A chord	The perpendicular from the centre. It bisects the chord.
An angle at the centre and one at the circumference, on the same arc	The centre one is double.
A diameter and a third point on the circle	That angle is a right angle.
Two angles standing on the same arc	They are equal.
A quadrilateral inscribed in a circle	Opposite angles sum to 180°.
A tangent and a chord meeting	That angle equals the angle in the alternate segment.
Two tangents from the same external point	They have equal length, and the line to the centre bisects the angle between them.
Two circles	The line of centres. Everything about tangency and common chords lives on it.

57.2 The circle in coordinates

Three habits that turn most coordinate circle questions into one line.

— **Complete both squares immediately.** $x^2+y^2+2gx+2fy+c=0$ becomes $(x+g)^2+(y+f)^2=g^2+f^2-c$, so the centre is $(-g,-f)$ and the radius is the square root of the right-hand side. If that right-hand side is negative there is no circle, which is a genuine question type.

— **The perpendicular bisector of any chord passes through the centre.** So a circle through three given points is found by intersecting two perpendicular bisectors, and a circle through two given points has its centre on one perpendicular bisector, which is usually enough with one more condition.

— **Tangency to a line is a distance condition**, not an algebra condition: the distance from the centre to the line equals the radius. That is faster than substituting and setting a discriminant to zero, and it never produces a quartic.

57.3 TWO CIRCLES, BY SUBTRACTION

CORE

TRIGGER Two circle equations, asked for their common chord, their intersections, or whether they meet at all.

CLAIM **Subtract the equations.** The x^2 and y^2 terms cancel and you are left with the straight line through both intersection points. Substitute back once. To test whether they meet, compare the distance d between the centres with $R + r$ and $|R - r|$: they meet twice if $|R - r| < d < R + r$, touch if d equals either, and miss otherwise.

WHY Any common point satisfies both equations and therefore their difference, and the difference of two circle equations is linear.

COST Thirty seconds against solving a pair of quadratics, which usually produces a quartic you cannot factor under pressure.

BREAKS Subtraction gives you a line **whether or not the circles actually intersect.** Check the distance between centres before believing the two points you find.

DEFAULT Substitute one equation into the other.

57.4 COMMON TANGENT LENGTHS

EDGE

TRIGGER Two circles, radii R and r, centres d apart, and a question about the length of a common tangent.

CLAIM External common tangent $= \sqrt{d^2 - (R - r)^2}$. Internal common tangent $= \sqrt{d^2 - (R + r)^2}$.

WHY Drop a parallel from the smaller centre to make a right-angled triangle whose legs are the tangent length and the difference, or the sum, of the radii. Then Pythagoras.

COST One line against constructing the configuration from scratch. Rare, and unmistakable when it appears.

BREAKS The internal tangent needs $d \geq R + r$, so if the circles overlap the formula returns the square root of a negative number, which is the answer to a different question.

DEFAULT Coordinates, with one centre at the origin.

57.5 POWER OF A POINT

EDGE

TRIGGER Two chords crossing inside a circle, or two secants from an external point, or a tangent and a secant from the same external point.

CLAIM From any point P, the product of the two distances along a line through the circle is the same for every such line, and equals the square of the tangent length when a tangent exists: $PA \cdot PB = PC \cdot PD = PT^2$.

WHY The two triangles formed are similar, because angles in the same segment are equal. That is the one-line proof and it is also the on-specification route.

COST	Turns a similar-triangles argument into one multiplication. Perhaps one question in four papers.
BREAKS	Distances **from** P along each line, in order. Using the chord length instead of the distance from P is the standard error.
DEFAULT	The similar triangles it came from. Always available, and on the specification, which power of a point is not.

DRILL A RECOGNITION. NAME THE FACT AND THE LINE YOU WOULD DRAW, TEN SECONDS EACH.

1. AB is a diameter and C is on the circle. What is $\angle ACB$?
2. A chord of length 10 in a circle of radius 13. What do you draw, and what do you get?
3. Two tangents from P touch at A and B. Name two facts.
4. $ABCD$ is cyclic with $\angle A = 105°$. Find $\angle C$.
5. A tangent at T and a chord TQ make an angle of $40°$. What else is $40°$?

ANSWERS. 1. $90°$. 2. The perpendicular from the centre; it bisects the chord, giving a right triangle with legs 5 and $\sqrt{169-25}=12$, so the chord is 12 from the centre. 3. $PA = PB$, and OP bisects $\angle APB$ (and is the perpendicular bisector of AB). 4. $75°$. 5. The angle in the alternate segment, that is the inscribed angle subtending TQ from the other arc.

DRILL B EXAM STANDARD.

1. Find the length of the common chord of $x^2 + y^2 = 25$ and $(x-8)^2 + y^2 = 25$.
2. A chord of length 16 sits in a circle of radius 10. How far is it from the centre?
3. Two circles have radii 9 and 4 and their centres are 13 apart. How long is their external common tangent, and what is special about this pair?
4. From a point P outside a circle, a tangent has length 12 and a secant through P meets the circle at distances 8 and x. Find x.
5. Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$, and the distance from its centre to the line $3x + 4y = 0$.

ANSWERS. 1. Subtracting gives $x = 4$, then $y^2 = 9$, so the chord has length 6. 2. $\sqrt{100-64}=6$. 3. $\sqrt{169-25}=12$; and since $d = R + r$ the circles touch externally, so there is no internal common tangent. 4. $144 = 8x$, so $x = 18$. 5. $(x-3)^2 + (y+2)^2 = 25$, centre $(3, -2)$, radius 5; distance $= |9-8|/\sqrt{9+16} = \frac{1}{5}$, so the line cuts the circle.

58 Lines, distances and reflections

Coordinate geometry that is not about circles. Most of it is one idea used four ways: **the shortest route is always the perpendicular one.**

58.1 The perpendicular principle

58.1 SHORTEST MEANS PERPENDICULAR CORE

TRIGGER	"The shortest distance from", "the closest point on", "the minimum distance between".

CLAIM	From a **point to a line**, the shortest segment is perpendicular to the line. From a **point to a curve**, the segment is perpendicular to the *tangent* at the closest point, that is, it lies along the normal. From a **line to a curve** that do not meet, the closest point on the curve is where the **tangent is parallel to the line**. From a **point to a circle**, the shortest distance is $\lvert d - r \rvert$ where d is the distance to the centre.
WHY	All four are the same statement. If the connecting segment were not perpendicular, you could slide along and shorten it, so at a minimum it must be.
COST	Converts a minimisation into a gradient condition, which is one equation instead of a calculus problem.
BREAKS	The circle version needs the point outside for $d - r$ and inside for $r - d$; the modulus handles both, but only if you take it. And on a curve there can be **several** perpendicular candidates: check them all and compare.
DEFAULT	Write the distance squared as a function of one variable and minimise it. Slower, always correct, and worth doing when the geometry is unclear.

Two consequences that recur.

— **The closest point on a line ℓ to a point P** is the foot of the perpendicular. Find it by writing the perpendicular through P (gradient $-1/m$) and intersecting.

— **The perpendicular distance from (x_0, y_0) to $ax + by + c = 0$** is $\dfrac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$. This is not named on the specification but it is one line from the foot-of-perpendicular calculation, and it is the fastest way to test a line against a circle. Treat it as EDGE: worth holding, never worth guessing at.

58.2 Reflecting a point in a line

58.2 REFLECT BY GOING TWICE AS FAR — EDGE

TRIGGER	"Find the image of P after reflection in the line ℓ", or a shortest-path question where a route must touch a line.
CLAIM	Drop the perpendicular from P to ℓ, find its foot F, and the image is $P' = 2F - P$ read coordinatewise. Equivalently, F is the midpoint of PP' and $PP' \perp \ell$ — those two conditions define the reflection and are the fastest way to *check* one.
WHY	A reflection preserves distance to the mirror and reverses the perpendicular direction.
COST	Three lines. The alternative, solving the two conditions simultaneously from scratch, is five or six.
BREAKS	Vertical and horizontal mirrors, where the gradient argument degenerates. Just swap or negate the relevant coordinate directly.
DEFAULT	Impose midpoint-on-the-line and perpendicularity as two simultaneous equations.

The reason this earns a card is the shortest-path family. *"A point on the line ℓ is joined to A and to B, both on the same side of ℓ. Where is the total distance least?"* Reflect B in ℓ to get B'; the shortest path is the straight line AB', and the answer is where it crosses ℓ. That is one construction, and without it the question is a calculus problem you do not have time for.

58.3 Angles, gradients and ratios

— **Gradient is the tangent of the angle of inclination.** If a line makes an angle θ with the positive x-axis then $m = \tan\theta$. So "the line makes an angle of $60°$" means $m = \sqrt{3}$, and a question about the angle *between* two lines is a question about their gradients.

— **Perpendicular gradients multiply to** -1, except when one line is vertical, which is the case the algebra loses and the option list remembers.

— **Dividing a segment in a ratio.** The point dividing AB in the ratio $\lambda : \mu$ from A is $\dfrac{\mu A + \lambda B}{\lambda + \mu}$. The midpoint is the case $1 : 1$. Note which letter goes with which weight: the weights are *swapped* relative to the ratio, which is the standard slip.

DRILL A RECOGNITION ONLY. NAME THE CONDITION, TEN SECONDS EACH.

1. Shortest distance from a point to a curve. What is the connecting segment perpendicular to?
2. Shortest distance from a line to a curve it misses. What condition locates the point?
3. A line makes $45°$ with the positive x-axis. Gradient?
4. P' is the reflection of P in ℓ. Name the two conditions it satisfies.

ANSWERS. 1. The tangent at the closest point, so it lies along the normal. 2. The tangent there is parallel to the line. 3. $\tan 45° = 1$. 4. The midpoint of PP' lies on ℓ, and PP' is perpendicular to ℓ.

DRILL B EXAM STANDARD.

1. Find the shortest distance from $(7, 3)$ to the line $y = 2x - 1$.
2. Find the shortest distance between the line $y = x - 6$ and the curve $y = x^2$.
3. Find the image of $(4, 1)$ after reflection in $y = 2x$.
4. $A = (1, 2)$ and $B = (9, 10)$. Find the point dividing AB in the ratio $3 : 1$ from A.

ANSWERS. 1. Line as $2x - y - 1 = 0$: distance $= |14 - 3 - 1|/\sqrt{5} = 10/\sqrt{5} = 2\sqrt{5}$. 2. Tangent parallel to the line means $2x = 1$, so $x = \frac{1}{2}$ and the point is $(\frac{1}{2}, \frac{1}{4})$; distance to $x - y - 6 = 0$ is $|\frac{1}{2} - \frac{1}{4} - 6|/\sqrt{2} = \frac{23}{4\sqrt{2}} = \frac{23\sqrt{2}}{8}$. 3. Perpendicular through $(4, 1)$ has gradient $-\frac{1}{2}$: $y = 1 - \frac{1}{2}(x - 4)$. Meeting $y = 2x$: $2x = 3 - \frac{x}{2}$, so $x = \frac{6}{5}$, $y = \frac{12}{5}$, and the image is $(\frac{12}{5} - 4,\ \frac{24}{5} - 1) = (-\frac{8}{5},\ \frac{19}{5})$. 4. $\dfrac{1 \cdot A + 3 \cdot B}{4} = (\frac{1+27}{4}, \frac{2+30}{4}) = (7, 8)$.

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59 Triangle machinery, priced honestly

This is where the first printing drifted furthest from the specification, so this section is mostly a price list, and the price list is the lesson.

59.1 What is actually named, and what it does

RESULT	TIER	HONEST VERDICT
Sine and cosine rules, and area $\frac{1}{2}ab\sin C$	CORE	Named in the specification. Includes the ambiguous case, which is examinable and was missing from the first printing.
Pythagorean families $3, 4, 5$; $5, 12, 13$; $8, 15, 17$; $7, 24, 25$	CORE	Recognition saves ten seconds a question and they recur. So do their multiples.
Equilateral triangle facts: height $\frac{\sqrt{3}}{2}x$**, area** $\frac{\sqrt{3}}{4}x^2$	CORE	Constant use, and the $1 : \sqrt{3} : 2$ ratio is the fastest way to rebuild them.

RESULT	TIER	HONEST VERDICT
Right triangle with its altitude to the hypotenuse	EDGE	Genuinely quick, one-line derivation, appears inside longer questions. §59.3.
Apollonius, the median-length formula	BEYOND	Put the midpoint at the origin and the whole thing is two lines of coordinates. Learning the formula is optional.
Stewart's theorem	BEYOND	A generalisation of a result you did not need.
Ceva and Menelaus	BEYOND	Not on the specification, no past question has needed them, and the configurations that trigger them do not occur on this test. Keep for olympiad.
The $3, 5, 7$ triangle with its 120° angle	BEYOND	A pleasing fact. Once every few years, if ever.

Read that table as a *method* rather than as a verdict. Anybody can produce a list of true results. The useful question is what each one costs to hold and what it returns, and that question has an answer you can compute. §74 is the arithmetic.

59.2 The ambiguous case, which is the examinable one

59.2 TWO SIDES AND A NON-INCLUDED ANGLE CAN GIVE TWO TRIANGLES — CORE

TRIGGER You are given two sides and an angle that is **not** between them, and asked for a missing angle or side, or asked how many triangles are possible.

CLAIM The sine rule gives $\sin\theta$, and $\sin\theta = \sin(180^\circ - \theta)$, so there are two candidate angles. Both are valid exactly when the second one still leaves a positive third angle. Concretely, with sides a, b and angle A opposite a: no triangle if $a < b\sin A$; exactly one if $a = b\sin A$ (right-angled) or if $a \geq b$; and **two** if $b\sin A < a < b$.

WHY The side of length a swung from one end can meet the far side in two places, which is the picture the algebra is describing.

COST It is the difference between the right answer and the answer that is on the option list.

BREAKS The cosine rule has no ambiguity, because $\cos$ is one-to-one on 0° to 180°. So when a route through the cosine rule exists, take it and the whole issue disappears.

DEFAULT Draw the triangle to rough scale and see whether the second position is geometrically possible.

That last line in BREAKS is the real content of the card. **Prefer the cosine rule whenever both are available**, because it cannot produce a spurious second answer. Choosing between two valid routes on the grounds of which one has fewer failure modes is exactly the selection skill Book One §15 is about.

59.3 The altitude to the hypotenuse

59.3 ONE ALTITUDE, THREE SIMILAR TRIANGLES — EDGE

TRIGGER A right-angled triangle with the perpendicular dropped from the right angle to the hypotenuse. Any question involving both a leg and a piece of the hypotenuse.

CLAIM With legs a and b, hypotenuse c, altitude h, and the hypotenuse split into m next to a and n next to b:

$$ab = ch, \qquad a^2 = mc, \qquad b^2 = nc, \qquad h^2 = mn.$$

WHY One altitude creates three similar triangles. Each identity is one similarity ratio written out, so if you can see the picture you can rebuild all four in seconds. Do not memorise four formulae, memorise the similarity.

COST Saves a Pythagoras and a solve on the questions where it appears. Perhaps one in three papers, usually inside a longer question.

BREAKS The right angle has to be at the vertex the altitude comes from. In any other triangle none of these hold.

DEFAULT Similar triangles, written out, or coordinates.

DRILL A RECOGNITION ONLY. NAME THE RULE, TEN SECONDS EACH.

1. Two sides and the angle between them, asked for the third side. Which rule?
2. Two sides and a non-included angle. Which rule, and what must you check?
3. Sides 7 and 9 with an included angle of $30°$. Area?
4. A triangle with sides 8, 15 and 17. What do you not need to compute?

ANSWERS. 1. The cosine rule. 2. The sine rule, and the ambiguous case (§59.2). Prefer the cosine rule whenever it is available, because it cannot produce a spurious second answer. 3. $\frac{1}{2} \cdot 7 \cdot 9 \cdot \frac{1}{2} = \frac{63}{4}$. 4. Anything: it is a Pythagorean triple, so the angle between 8 and 15 is a right angle.

DRILL B EXAM STANDARD.

1. A triangle has $b = 8$, $c = 6$ and $A = 60°$. Find a and the area.
2. A triangle has $a = 5$, $b = 7$ and $A = 30°$. How many such triangles are there?
3. In a right-angled triangle the altitude to the hypotenuse divides it into pieces of length 4 and 9. Find the altitude and both legs.
4. An equilateral triangle has area $9\sqrt{3}$. Find its side and its height.

ANSWERS. 1. $a^2 = 64 + 36 - 2 \cdot 48 \cdot \frac{1}{2} = 52$, so $a = 2\sqrt{13}$; area $= \frac{1}{2} \cdot 8 \cdot 6 \cdot \frac{\sqrt{3}}{2} = 12\sqrt{3}$. 2. $b \sin A = 3.5$ and $b = 7$, and $3.5 < 5 < 7$, so **two**. 3. $h = \sqrt{4 \cdot 9} = 6$; $c = 13$, so $a^2 = 4 \cdot 13 = 52$ and $b^2 = 9 \cdot 13 = 117$, giving $a = 2\sqrt{13}$ and $b = 3\sqrt{13}$. 4. $\frac{\sqrt{3}}{4}x^2 = 9\sqrt{3}$ gives $x^2 = 36$, $x = 6$, and height $3\sqrt{3}$.

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60 Awkward areas, and how to decompose them

Almost every hard-looking area question on this paper is a subtraction. The skill is choosing the two shapes.

60.1 The decomposition habit

60.1 NAME A SHAPE YOU KNOW, THEN SUBTRACT CORE

TRIGGER A shaded region bounded by arcs, chords, or several straight edges, with no formula that fits it.

CLAIM Every such region is a signed sum of shapes you already have: triangles, rectangles, sectors, segments, and areas under curves. Write the region as a sum and difference of those before computing anything.

WHY	Area is additive. That is the entire justification, and it is why the method never fails.
COST	It is the intended route. The alternative is inventing a formula for a shape nobody has a formula for.
BREAKS	Overlaps counted twice, and pieces outside the region counted once. Draw the decomposition on the board and shade what you are adding and what you are removing.
DEFAULT	Coordinates and integration, when the boundary is a curve you can write down.

THE FOUR DECOMPOSITIONS THAT COVER ALMOST EVERYTHING

— **Segment = sector − triangle.** The single most common one, and the reason §53.3 tells you to derive rather than memorise.

— **Region between touching circles = the polygon joining the centres − the sectors inside it.** For three touching circles of radius r the polygon is an equilateral triangle of side $2r$, and the three sectors make exactly a half-circle because a triangle's angles sum to $180°$. Area $\sqrt{3}r^2 - \frac{1}{2}\pi r^2$.

— **Inscribed and circumscribed shapes:** work through the *radius*. Square in a circle, the diagonal is the diameter; circle in a square, the side is. For an equilateral triangle the circumradius is $\frac{2}{3}$ of the height and the inradius $\frac{1}{3}$. The wrong way round is the designed error.

— **Region between a curve and a chord = the trapezium under the chord − the area under the curve**, or the identity in §54 if it is a polynomial and a line.

60.2 Two small results that keep appearing

— **A rhombus, or any quadrilateral with perpendicular diagonals, has area $\frac{1}{2}d_1d_2$.** Because the diagonals cut it into four right triangles that reassemble into a rectangle of half the size.

— **A triangle with a vertex at the origin** and other vertices (x_1, y_1), (x_2, y_2) has area $\frac{1}{2}\,|x_1y_2 - x_2y_1|$. EDGE: not named on the specification, one line from the standard formula, and it turns a coordinate area question into one multiplication. Translate a vertex to the origin first if none is there.

DRILL A RECOGNITION ONLY. NAME THE DECOMPOSITION, TEN SECONDS EACH.

1. The region between a chord and its arc.
2. Three mutually touching circles of radius r. Which polygon do you draw?
3. A square inscribed in a circle of radius r. What is the side?
4. A quadrilateral with perpendicular diagonals of 6 and 10. Area?

ANSWERS. 1. Sector minus triangle. 2. The equilateral triangle joining the three centres, of side $2r$. 3. The diagonal is the diameter $2r$, so the side is $r\sqrt{2}$. 4. $\frac{1}{2} \cdot 6 \cdot 10 = 30$.

DECOMPOSE, OR GO TO COORDINATES?

Both routes always work, so the question is which is cheaper, and you can settle it in five seconds.

Decompose	The boundary is made of arcs, chords and straight edges, and every piece is a shape with a formula. Angles are given, or fall out of §57.1. This is almost always the cheaper route on a plane-geometry question.
Go to coordinates	The boundary involves a curve you can write down, the figure is already given with coordinates, or you cannot see the angles. Slower, and it never lies.

The tell is what you are given. **Angles and radii point at decomposition; points and equations point at coordinates.** If you have been given neither, you have probably missed a condition, which is Book One §7.

DRILL B EXAM STANDARD.

1. Three circles of radius 2 touch one another. Find the area of the region enclosed between them.
2. A square of side 4 has a quarter circle of radius 4 drawn from one corner. Find the area outside the quarter circle but inside the square.
3. A regular hexagon is inscribed in a circle of radius 6. Find the area between them.
4. Find the area of the triangle with vertices $(1, 1)$, $(5, 2)$ and $(3, 7)$.

ANSWERS. 1. $\sqrt{3} \cdot 4 - \frac{1}{2}\pi \cdot 4 = 4\sqrt{3} - 2\pi$. 2. $16 - \frac{1}{4}\pi \cdot 16 = 16 - 4\pi$. 3. The hexagon is six equilateral triangles of side 6, area $6 \cdot \frac{\sqrt{3}}{4} \cdot 36 = 54\sqrt{3}$; the circle is 36π; the difference is $36\pi - 54\sqrt{3}$. 4. Translate by $(-1, -1)$: vertices $(0, 0)$, $(4, 1)$, $(2, 6)$; area $= \frac{1}{2}|4 \cdot 6 - 2 \cdot 1| = 11$.

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PART XIII

NUMBER, SEQUENCES AND COUNTING

The most consistently under-prepared part of the paper, and the raw material of half of Paper 2, because a claim about all integers is the easiest kind of claim to set and to break.

61 Sequences and series: the traps that get set

62 Sigma, telescoping and the method of differences

63 Recursions: solve, iterate, or spot the loop

64 Number, divisibility and counting

65 Selections and arrangements

66 Diophantine problems

§64 is worth more than it looks. Parity and residues settle two or three questions a sitting and cost ten seconds each.

61 Sequences and series: the traps that get set

The three-term conditions and index-hopping are well known. The things that actually go wrong are not.

61.1 The definitions, once, so the rest is short

Arithmetic	$a_n = a + (n-1)d$ and $S_n = \frac{n}{2}\big(2a + (n-1)d\big) = \frac{n}{2}(a_1 + a_n)$.
Geometric	$b_n = ar^{n-1}$ and $S_n = \frac{a(1-r^n)}{1-r}$ for $r \neq 1$, with $S_\infty = \frac{a}{1-r}$ when $\lvert r\rvert < 1$.

Two conditions hiding in that table. The geometric sum formula **requires** $r \neq 1$, and the sum to infinity **requires** $|r| < 1$. When r contains a parameter, the question is very often about the range of the parameter and not about the sum at all.

61.2 Three terms, two conditions

61.2 DOUBLE THE MIDDLE, SQUARE THE MIDDLE — CORE

TRIGGER Three consecutive terms named, or three terms described as being in arithmetic or geometric progression.

CLAIM Arithmetic: $2a_{n+1} = a_n + a_{n+2}$, double the middle is the sum of the outer two. Geometric: $b_{n+1}^2 = b_n b_{n+2}$, square the middle is the product of the outer two. Both survive **hopping**, as long as the hops are equal: $2a_n = a_{n-k} + a_{n+k}$ and $b_n^2 = b_{n-k}b_{n+k}$.

WHY The common difference between equally spaced terms is kd either side, and the common ratio is r^k either side.

COST Very large. It replaces five unknowns, a, d, b, r and the terms, with one equation. A question with p, q, p^2 in arithmetic progression and p, p^2, q in geometric progression is two lines this way and half a page the other way.

BREAKS Consecutive or **equally spaced** only. Check the indices add up: $2n = (n-k) + (n+k)$.

DEFAULT Write $a_n = a + (n-1)d$ and solve.

61.3 Sum and term, in both directions

61.3 THE SUM OF AN ARITHMETIC SEQUENCE IS A QUADRATIC WITH NO CONSTANT TERM — CORE

TRIGGER You have the nth term of an arithmetic sequence and want the sum, or you have S_n and want the terms.

CLAIM For $a_n = pn + q$, $S_n = \frac{p}{2}n^2 + \left(q + \frac{p}{2}\right)n$, with **no constant term**. Backwards, from S_n: $a_n = S_n - S_{n-1}$ for $n \geq 2$, and $a_1 = S_1$ separately.

WHY $S_n = \frac{n}{2}(a_1 + a_n)$, expanded. There is no constant term because $S_0 = 0$.

COST Ten seconds against expanding a sigma sum. It is also the cleanest check available: the coefficients of S_n must sum to a_1.

BREAKS The $n \geq 2$ in the backwards direction. Every year somebody computes a_1 from $S_1 - S_0$ using a formula that was never valid at $n = 1$, and the wrong answer is on the option list.

DEFAULT $\sum(pk + q) = p \cdot \dfrac{n(n+1)}{2} + qn.$

The converse is a useful recognition cue in its own right: **if S_n is a quadratic in n with no constant term, the sequence is arithmetic**, and the common difference is twice the leading coefficient.

61.4 Nested shapes, and the ratio people get wrong

A square, or a triangle, or a circle, nested inside the previous one for ever, with a total length or a total area asked for.

— Find the **length** ratio r first. That is the geometric step, and it is usually a similarity or a Pythagoras.

— Lengths then form a geometric series with ratio r; **areas** form one with ratio r^2; volumes with r^3.

— The total area is $\dfrac{A_1}{1 - r^2}$, which converges whenever $|r| < 1$.

Trying to find the area ratio directly is where this goes wrong, because the area ratio is not usually something you can see. The length ratio always is.

DRILL A RECOGNITION ONLY. ONE EQUATION EACH, TEN SECONDS.

1. p, q, r are consecutive terms of an arithmetic progression. Write one equation.
2. p, q, r are consecutive terms of a geometric progression. Write one equation.
3. $S_n = 2n^2 + 3n$. Is the sequence arithmetic, and how do you know at a glance?
4. Squares nested with side ratio $\frac{2}{3}$. What is the ratio for the areas?

ANSWERS. 1. $2q = p + r$. 2. $q^2 = pr$. 3. Yes: a quadratic in n with no constant term is exactly the shape of an arithmetic sum. 4. $\frac{4}{9}$. Find the length ratio, then square it.

DRILL B EXAM STANDARD.

1. The first three terms of an arithmetic progression are p, q, p^2, and the first three terms of a geometric progression are p, p^2, q, with $p < 0$. Find the sum of the first ten terms of the arithmetic progression.
2. $S_n = 3n^2 - n$ for a sequence. Find a_1 and a_{10}, and say why the sequence is arithmetic.
3. A square of side 1 has a square inscribed by joining the midpoints of its sides, and so on for ever. Find the total area of all the squares.
4. A geometric series has first term 5 and sum to infinity 4. Find r, or explain why no such series exists.

ANSWERS. 1. $2q = p + p^2$ and $(p^2)^2 = pq$ give $q = p^3$, so $2p^3 = p + p^2$, that is $p(2p+1)(p-1) = 0$; with $p < 0$ this leaves $p = -\frac{1}{2}$ and $q = -\frac{1}{8}$. Then $d = q - p = \frac{3}{8}$ and $S_{10} = 10\left(-\frac{1}{2}\right) + 45\left(\frac{3}{8}\right) = -5 + \frac{135}{8} = \frac{95}{8}$. 2. $a_1 = S_1 = 2$; $a_{10} = S_{10} - S_9 = 290 - 234 = 56$; S_n is a quadratic in n with no constant term, which is exactly the shape of an arithmetic sum. 3. Side ratio $1/\sqrt{2}$, so area ratio $\frac{1}{2}$, and the total is $1/\left(1 - \frac{1}{2}\right) = 2$. 4. $5/(1 - r) = 4$ gives $1 - r = \frac{5}{4}$, so $r = -\frac{1}{4}$, and $|r| < 1$, so it exists.

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62 Sigma, telescoping and the method of differences

Sigma notation is not a topic. It is a way of writing things down, and almost every difficulty with it is a difficulty about **which terms are included**.

62.1 Reading a sigma sum

- **Count the terms first.** $\sum_{k=m}^{n}$ has $n-m+1$ terms. Forgetting the $+1$ is the most common slip in the whole of Part XIII.
- **Anything without a k in it is a constant**, so $\sum_{k=1}^{n} 7 = 7n$ and $\sum_{k=1}^{n} f(x) = nf(x)$.
- **Split at will:** $\sum_{k=m}^{n} = \sum_{k=1}^{n} - \sum_{k=1}^{m-1}$, and sums add termwise.
- **Shift the index deliberately**, not by accident. Replacing k by $k+1$ shifts both limits down by one, and the two must move together.

The standard sums, in decreasing order of how often you need them:

$$\sum_{k=1}^{n} 1 = n, \qquad \sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \qquad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2.$$

The first two are CORE. The last two are EDGE: they are quotable, they occasionally save a minute, and if you cannot recall one under pressure the honest move is to sum the first three or four values and fit rather than to guess a formula.

62.2 The method of differences

62.2 IF EACH TERM IS A DIFFERENCE, THE SUM COLLAPSES — EDGE

TRIGGER A sum whose terms are fractions with factorised denominators, like $\dfrac{1}{k(k+1)}$, or a sum you are told to evaluate that has no standard formula, or the words "method of differences".

CLAIM If you can write $u_k = v_k - v_{k+1}$, then $\sum_{k=1}^{n} u_k = v_1 - v_{n+1}$. Everything in the middle cancels in pairs. For $\dfrac{1}{k(k+1)}$ the split is $\dfrac{1}{k} - \dfrac{1}{k+1}$, so the sum is $1 - \dfrac{1}{n+1} = \dfrac{n}{n+1}$.

WHY Write out the first three terms and the last two, and look at what survives. The proof *is* the writing out.

COST It converts a sum you cannot do into two terms. When it applies, nothing else comes close.

BREAKS Terms that cancel two apart rather than one apart, like $\dfrac{1}{k(k+2)}$, where **two** terms survive at each end rather than one. Always write out the first three and the last three before committing to the answer.

DEFAULT Compute the sum for $n = 1, 2, 3$, guess the closed form, and check at $n = 4$. Slower, and it works on sums you cannot split.

THE TWO SPLITS WORTH RECOGNISING ON SIGHT

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}, \qquad \frac{1}{k(k+2)} = \frac{1}{2}\left(\frac{1}{k} - \frac{1}{k+2}\right).$$

Both come from the same one-line calculation: write $\dfrac{1}{k(k+c)} = \dfrac{A}{k} + \dfrac{B}{k+c}$, multiply up, and compare. You get $A = 1/c$ and $B = -1/c$ every time, so the general split is $\dfrac{1}{k(k+c)} = \dfrac{1}{c}\left(\dfrac{1}{k} - \dfrac{1}{k+c}\right)$.

Deriving it in ten seconds is better than remembering it, because the derivation also tells you where the $\frac{1}{2}$ came from.

62.3 Interleaved and alternating sums

Two shapes that look harder than they are.

— **Interleaved.** A sum whose odd-numbered and even-numbered terms follow different rules. Split it into two sums, each with its own formula, and add. The only care needed is counting how many terms each half has, which depends on the parity of n — and "in terms of n" answers of this kind usually need two cases.

— **Alternating.** A sum like $1 - 2 + 3 - 4 + \cdots$. **Pair the terms** from the front. Each pair contributes the same amount, and if the number of terms is odd there is one left over. That is faster and safer than trying to find a closed form.

DRILL A RECOGNITION ONLY. TEN SECONDS EACH.

1. How many terms are there in $\displaystyle\sum_{k=7}^{31}$?
2. $\displaystyle\sum_{k=1}^{n} 5$?
3. $\dfrac{1}{k(k+3)}$. What is the split?
4. $1 - 2 + 3 - \cdots - 40$. What is the move?

ANSWERS. 1. $31 - 7 + 1 = 25$. 2. $5n$: nothing in it depends on k. 3. $\frac{1}{3}\left(\frac{1}{k} - \frac{1}{k+3}\right)$, and note that **three** terms survive at each end, not one. 4. Pair from the front: twenty pairs each contributing -1, so -20.

DRILL B EXAM STANDARD.

1. Evaluate $\displaystyle\sum_{k=1}^{20}(3k - 2)$.
2. Evaluate $\displaystyle\sum_{k=1}^{n}\frac{1}{k(k+1)}$.
3. Evaluate $1 - 2 + 3 - 4 + \cdots + 99 - 100$.
4. Evaluate $\displaystyle\sum_{k=5}^{30} k$.

ANSWERS. 1. Arithmetic with $a_1 = 1$, $a_{20} = 58$, twenty terms: $\frac{20}{2}(1 + 58) = 590$. 2. Telescoping: $1 - \frac{1}{n+1} = \frac{n}{n+1}$. 3. Fifty pairs each contributing -1, so -50. 4. $\frac{30\cdot 31}{2} - \frac{4\cdot 5}{2} = 465 - 10 = 455$. Check the term count: $30 - 5 + 1 = 26$ terms.

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63 Recursions: solve, iterate, or spot the loop

A recurrence gives you the next term from the current one. There are exactly three things a TMUA question can want, and telling them apart in five seconds is the section.

63.1 Absurd index means find the loop

63.1 COMPUTE TERMS UNTIL ONE REPEATS

CORE

TRIGGER	A recurrence $x_{n+1} = f(x_n)$ and a question asking for a term with an absurd index: the 100th, the 2026th.
CLAIM	Compute terms until one repeats. The sequence is then periodic with that period p, and $x_N = x_r$ where r is N reduced modulo p into the range of your first block.
WHY	A recurrence of this form is determined entirely by its current value, so the moment a value recurs the whole future recurs.
COST	The absurd index is the cue and it is unmistakable. Thirty seconds.
BREAKS	**Indexing.** Reducing 2026 modulo 3 gives 1, and whether that means x_1 or x_4 depends on where your sequence starts. Write out x_1 to x_4 and match; do not trust the arithmetic alone.
DEFAULT	There is no fallback. Computing 2026 terms is not a plan.

63.2 Linear recurrence: find the fixed point

63.2 SHIFT TO THE FIXED POINT AND IT BECOMES GEOMETRIC

EDGE

TRIGGER	$x_{n+1} = ax_n + b$ with $a \neq 1$, and a question about a general term or a limit.
CLAIM	Solve $L = aL + b$, so $L = \dfrac{b}{1-a}$. Then $x_n - L = a^n(x_0 - L)$, which gives the closed form $x_n = L + a^n(x_0 - L)$. If $\lvert a\rvert < 1$ the sequence converges to L; if $\lvert a\rvert > 1$ it diverges; if $a = -1$ it oscillates between two values.
WHY	Subtract the fixed-point equation from the recurrence: $x_{n+1} - L = a(x_n - L)$. What is left is a geometric sequence in the displacement from L.
COST	Turns "find the limit" and "find the nth term" into one line each. It also explains *why* the limit is what it is.
BREAKS	$a = 1$, where there is no fixed point and the sequence is arithmetic instead, with $x_n = x_0 + nb$. Check that case before dividing by $1 - a$.
DEFAULT	Compute five terms and look for the pattern.

THE LIMIT ARGUMENT THAT IS ONLY HALF AN ARGUMENT

"If $x_n \to L$ then $L = f(L)$" is true, and it is the standard way to find a limit. But it assumes the limit exists, and Paper 2 sets questions on exactly that gap.

$x_{n+1} = 2x_n$ with $x_1 = 1$ gives $L = 2L$, so $L = 0$, which is nonsense: the sequence runs away to infinity. The fixed-point equation was solved correctly and the conclusion was still wrong, because the sequence has no limit to find.

So the honest statement is: **solve $L = f(L)$ to find the only possible limits, then justify separately that the sequence converges** — usually by showing it is monotonic and bounded, or by the $\lvert a\rvert < 1$ argument above. On Paper 1 that second half is often assumed; on Paper 2 it is the question.

63.3 Neither of the above

If the recurrence is not linear and does not loop, the intended route is almost always to **compute the first four or five terms and look for structure**: a pattern in the numerators and denominators, a product that telescopes, a quantity that stays constant. That last one is the most valuable habit in the section: look for an *invariant*, something built from x_n that does not change from term to term. If you find one, the problem is finished.

DRILL A RECOGNITION ONLY. NAME THE MOVE, TEN SECONDS EACH.

1. $x_{n+1} = f(x_n)$, asked for x_{500}. First move?
2. $x_{n+1} = \frac{1}{3}x_n + 4$. What is the fixed point?
3. The same recurrence: does it converge, and why?
4. $x_{n+1} = x_n + 7$. Why does the fixed-point method fail here?

ANSWERS. 1. Compute terms until one repeats, then reduce the index modulo the period. 2. $L = 4/(1 - \frac{1}{3}) = 6$. 3. Yes, because $|a| = \frac{1}{3} < 1$, so the displacement from L is geometric and shrinks. 4. $a = 1$, so there is no fixed point; the sequence is arithmetic, $x_n = x_0 + 7n$.

DRILL B EXAM STANDARD.

1. $x_1 = 2$ and $x_{n+1} = \dfrac{1}{1 - x_n}$. Find x_{2026}.
2. $x_1 = 5$ and $x_{n+1} = \frac{1}{2}x_n + 3$. Find the limit, and the closed form for x_n.
3. $x_1 = 1$ and $x_{n+1} = \dfrac{x_n}{1 + x_n}$. Find x_{100}.
4. $x_{n+1} = 3x_n - 4$ with $x_1 = 2$. Does the sequence converge? What if $x_1 = 3$?

ANSWERS. 1. The terms run $2, -1, \frac{1}{2}, 2, \ldots$, so the period is 3; $2026 = 3 \cdot 675 + 1$, so $x_{2026} = x_1 = 2$. 2. $L = \frac{1}{2}L + 3$ gives $L = 6$; $x_n - 6 = \left(\frac{1}{2}\right)^{n-1}(5 - 6)$, so $x_n = 6 - \left(\frac{1}{2}\right)^{n-1}$, and the limit is 6. 3. Neither periodic nor linear, so look for an invariant. Taking reciprocals, $\dfrac{1}{x_{n+1}} = \dfrac{1 + x_n}{x_n} = \dfrac{1}{x_n} + 1$, so $1/x_n$ is arithmetic with first term 1 and common difference 1, giving $1/x_n = n$ and $x_{100} = \frac{1}{100}$. 4. $L = 3L - 4$ gives $L = 2$, and $x_1 = 2$ already, so the sequence is constant and converges trivially. With $x_1 = 3$: $x_n - 2 = 3^{n-1}(3 - 2) = 3^{n-1}$, and $|a| = 3 > 1$, so it diverges. Same fixed point, opposite behaviour — which is why "$L = f(L)$" is only half an argument.

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64 Number, divisibility and counting

The most consistently neglected area on the specification, and the raw material of half of Paper 2.

64.1 Parity and residues as an eliminator

64.1 TWO CASES, TEN SECONDS CORE

TRIGGER — Any claim or equation about integers, especially with squares in it, and any option list of integers.

CLAIM — Squares are 0 or 1 modulo 4, and 0 or 1 modulo 3. Products of consecutive integers carry guaranteed factors: $n(n+1)$ is even and $n(n+1)(n+2)$ is divisible by 6. Odd plus odd is even; odd times odd is odd; an odd number is never a difference of two even ones.

WHY — Write $n = 2k$ or $n = 2k + 1$ and square. Two cases, ten seconds, and it is the standard proof by cases the specification asks you to be able to construct.

COST	On Paper 1 it kills options without solving. On Paper 2 it produces the counterexample or the proof directly. Two or three questions a sitting reward it.
BREAKS	Nothing, but **choose the modulus deliberately.** Modulo 4 settles sums of two squares, modulo 3 settles most things with cubes, modulo 9 settles digit sums. Pick the one that makes the target trivial.
DEFAULT	Test small cases until the pattern appears, which is slower but always available.

The divisibility tests worth owning cold, because they turn a factorisation question into a glance:

2, 5, 10	Last digit.
4	Last two digits. **8:** last three.
3, 9	Digit sum.
11	Alternating digit sum.
6	Divisible by 2 and by 3. Similarly 12, 15, and any product of coprime factors.

64.2 Divisor counting

64.2 EXPONENTS, PLUS ONE, MULTIPLIED — CORE

TRIGGER	"How many factors", "how many divisors are squares", "the smallest k such that kn is a perfect square", or a question about the number of ways to write n as a product.
CLAIM	If $n = p_1^{e_1}p_2^{e_2}\cdots$ then n has $(e_1+1)(e_2+1)\cdots$ positive divisors. n is a perfect square exactly when every e_i is even, a cube when every e_i is a multiple of three. To make n a square, multiply by the primes with odd exponent.
WHY	A divisor is a choice of exponent for each prime, independently, from 0 to e_i. That is the product rule.
COST	Twenty seconds against listing factors in pairs, which is both slow and where people miss the square root.
BREAKS	Only **positive** divisors are counted. If the question says integer divisors, double it.
DEFAULT	Pair the factors up from 1 outwards and stop at $\sqrt{n}$.

Two relatives that come free from the same prime factorisation. **The HCF takes the smaller exponent of each prime and the LCM takes the larger**, from which $\mathrm{HCF}(a,b) \times \mathrm{LCM}(a,b) = ab$ falls out immediately, because each prime contributes $\min + \max =$ the sum of the two exponents.

64.3 The counting protocol

64.3 NAME THE CASES BEFORE YOU COUNT ANYTHING — CORE

TRIGGER	How many integers, how many ways, how many pairs, how many arrangements satisfy something.
CLAIM	Do not improvise. Four steps, always the same. **Name the cases so they cannot overlap. Count each case with the product rule. Check the two boundaries and the empty case. Add.**

WHY	Unstructured counting fails in exactly two ways, double counting and missing an endpoint, and both are eliminated by naming the cases before counting anything.
COST	This is the difference between a reliable mark and a coin flip on one or two questions a paper.
BREAKS	Cases that look disjoint and are not. If a case split needs the word "or", check whether anything satisfies both halves. The number of integers in $[a, b]$ is $\lfloor b \rfloor - \lceil a \rceil + 1$, and forgetting the $+1$ is the most common single slip in this area.
DEFAULT	List them. On a question with a small answer this is not defeat, it is the fastest route.

DRILL A RECOGNITION ONLY. TEN SECONDS EACH.

1. What values can n^2 take modulo 4?
2. How many positive divisors does $2^3 \cdot 5^2$ have?
3. Is $n(n+1)(n+2)$ always divisible by 6?
4. Smallest positive k making $50k$ a perfect square?

ANSWERS. 1. 0 or 1, never 2 or 3. 2. $4 \cdot 3 = 12$. 3. Yes: three consecutive integers contain a multiple of 2 and a multiple of 3. 4. $50 = 2 \cdot 5^2$, so the odd exponent is on the 2, giving $k = 2$.

DRILL B EXAM STANDARD.

1. How many positive divisors does $2^4 \cdot 3^3 \cdot 5$ have, and how many of them are perfect squares?
2. How many pairs of positive integers (a, b) satisfy $ab = 72$ with $a < b$?
3. Show that $n^2 + n + 1$ is odd for every integer n.
4. Find $\mathrm{HCF}(84, 360)$ and $\mathrm{LCM}(84, 360)$.

ANSWERS. 1. $5 \cdot 4 \cdot 2 = 40$ divisors; squares need even exponents, so $3 \cdot 2 \cdot 1 = 6$. 2. 72 has 12 divisors and is not a square, so 6 pairs. 3. $n^2 + n = n(n+1)$ is a product of consecutive integers, so even, so the whole thing is odd. 4. $84 = 2^2 \cdot 3 \cdot 7$ and $360 = 2^3 \cdot 3^2 \cdot 5$; HCF $= 2^2 \cdot 3 = 12$, LCM $= 2^3 \cdot 3^2 \cdot 5 \cdot 7 = 2520$. Check: $12 \times 2520 = 30240 = 84 \times 360$.

DRILL C THE 9.0 LAYER.

1. Prove or disprove: if n^2 is divisible by 12 then n is divisible by 12.
2. Find the smallest positive integer k for which $756k$ is a perfect cube.
3. How many integers between 1 and 300 are divisible by 3 or 5 but not by 15?
4. What is the smallest positive integer with exactly 12 positive divisors?

ANSWERS. 1. **False.** $n = 6$ gives $n^2 = 36$, divisible by 12, and 6 is not. Note that both jobs of a counterexample were checked, see §69. 2. $756 = 2^2 \cdot 3^3 \cdot 7$, so we need $2^1 \cdot 7^2 = 98$. 3. Multiples of 3: 100. Of 5: 60. Of 15: 20. Divisible by 3 or 5 is $100 + 60 - 20 = 140$, and removing the multiples of 15 leaves 120. 4. $12 = 12, 6 \cdot 2, 4 \cdot 3, 3 \cdot 2 \cdot 2$; the candidates are $2^{11} = 2048$, $2^5 \cdot 3 = 96$, $2^3 \cdot 3^2 = 72$, $2^2 \cdot 3 \cdot 5 = 60$. Smallest is **60**.

65 Selections and arrangements

AN HONEST NOTE ON TIER

The specification names **systematic listing** and the **binomial expansion** of $(1 + x)^n$ and $(a + f(x))^n$ for positive integer n. It does not name permutations and combinations as a topic in their own right.

> What that means in practice: the binomial expansion and its coefficients are CORE, and the counting machinery below is EDGE. It is not extra syllabus, it is a faster way to do the systematic listing the paper does ask for. Learn it as a shortcut with a default (list them), not as a topic.

65.1 The three counting moves

MOVE	WHEN AND HOW
Product rule	Independent choices multiply. r decisions with n_i options each give $n_1 n_2 \cdots n_r$ outcomes. This is the whole of counting; everything below is a special case.
Arrangements	n distinct objects in a line: $n!$. With repeats, divide by the factorial of each repeat count: the letters of BANANA give $\dfrac{6!}{3!\,2!} = 60$.
Selections	Choosing r from n without order: $\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$. Note $\binom{n}{r} = \binom{n}{n-r}$, which halves the arithmetic, and $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$, which is Pascal's triangle.

Two constraint patterns cover most questions that look harder.

— **Keep objects together:** glue them into one block, arrange the blocks, then arrange within the block. Three particular people together among n gives $(n-2)! \times 3!$.

— **Keep objects apart:** arrange everything else first, then place the awkward objects into the **gaps**. Arranging m others creates $m+1$ gaps, and you choose which to use. This is much safer than subtracting the "together" cases, which double-counts as soon as there are three or more objects to separate.

65.2 The binomial expansion

65.2 THE GENERAL TERM IS THE QUESTION — CORE

TRIGGER "Find the coefficient of x^k in", or an expansion with a large power you are clearly not meant to write out.

CLAIM $(a+b)^n = \displaystyle\sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$, so the term containing b^r is $\binom{n}{r} a^{n-r} b^r$. To find a particular power of x, set up the exponent equation and solve for r **before** computing anything.

WHY Each term of the expansion is one way of choosing b from r of the n brackets and a from the rest.

COST One binomial coefficient instead of an expansion. It is the intended route.

BREAKS When b itself contains a power of x, as in $\left(x^2 + \frac{1}{x}\right)^{12}$, the exponent equation is not $r = k$ and getting it wrong gives an answer that is on the option list. Write the exponent of x as a function of r, set it equal to the target, and check that r comes out a non-negative integer $\leq n$.

DEFAULT Expand the first few terms and look for the pattern.

65.3 Inclusion and exclusion

For two sets, $|A \cup B| = |A| + |B| - |A \cap B|$. For three, add the singles, subtract the pairs, add back the triple. The pattern continues, alternating.

The practical form on this paper is almost always: *how many integers up to N are divisible by a or by b?* Count $\lfloor N/a \rfloor$ and $\lfloor N/b \rfloor$, then subtract $\lfloor N/\mathrm{lcm}(a,b) \rfloor$. Using ab instead of the lcm is the standard error and it is wrong whenever a and b share a factor.

A two-way table is the same idea drawn out, and for two conditions it is usually faster and much harder to get wrong. Draw it whenever the question involves two yes-or-no properties.

DRILL A **RECOGNITION ONLY. NAME THE COUNT, TEN SECONDS EACH.**

1. Arrangements of the letters of LEVEL?
2. $\binom{9}{7}$, without touching 9!?
3. Five in a row, two of whom insist on sitting together. Which technique?
4. Coefficient of x^3 in $(1+x)^{10}$?

ANSWERS. 1. $\dfrac{5!}{2!\,2!} = 30$. 2. $\binom{9}{7} = \binom{9}{2} = 36$. 3. Glue them into one block: $4! \times 2!$. 4. $\binom{10}{3} = 120$.

DRILL B **EXAM STANDARD.**

1. How many arrangements are there of the letters of MISSISSIPPI?
2. In how many ways can 5 people sit in a row if two particular people must not sit together?
3. Find the coefficient of x^6 in $\left(x^2 + \dfrac{2}{x}\right)^{12}$.
4. How many integers from 1 to 200 are divisible by 4 or by 6?

ANSWERS. 1. $\dfrac{11!}{4!\,4!\,2!} = 34650$. 2. Total $5! = 120$; together, glue them: $4! \times 2! = 48$; apart: $120 - 48 = 72$. Or by gaps: arrange the other three in $3! = 6$ ways, creating 4 gaps, and choose 2 of them in order: $6 \times 4 \times 3 = 72$. 3. The general term is $\binom{12}{r}(x^2)^{12-r}(2/x)^r = \binom{12}{r}2^r x^{24-3r}$; setting $24 - 3r = 6$ gives $r = 6$, so the coefficient is $\binom{12}{6}2^6 = 924 \times 64 = 59136$. 4. $\lfloor 200/4 \rfloor = 50$, $\lfloor 200/6 \rfloor = 33$, lcm $= 12$ so $\lfloor 200/12 \rfloor = 16$; total $50 + 33 - 16 = 67$.

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66 Diophantine problems

Equations to be solved in integers. They look open-ended and they almost never are, because integrality is an enormously strong constraint.

66.1 Factorise, then read off the factor pairs

66.1 GET IT INTO THE FORM (SOMETHING)(SOMETHING) = A NUMBER EDGE

TRIGGER An equation in two integer unknowns, especially one containing a product term xy alongside x and y.

CLAIM Rearrange into a product equal to a constant, then list the factor pairs of that constant, **including the negative ones** if the variables are allowed to be negative. For $xy + ax + by = c$, add ab to both sides: $(x+b)(y+a) = c + ab$.

WHY Over the integers, a product with a fixed value has only finitely many factorisations, so an infinite-looking problem becomes a short list.

COST It is the intended route and there is rarely another one. Two minutes of aimless substitution becomes thirty seconds.

BREAKS Missing the negative factor pairs, and forgetting to check each candidate against any positivity condition in the question. Both are exactly what the distractors are built from.

DEFAULT Make one variable the subject, $y = \dfrac{\text{something}}{\text{something}}$, and ask when the division is exact. Slower, and it works when the factorisation does not.

66.2 Three ways to cut the search down

1. **Bound it.** If x and y are positive and $x \le y$, then x cannot exceed some small value and you check those one by one. Ordering the variables is free and removes the duplicates automatically.
2. **Use parity or a residue.** $x^2 + y^2 = 4k + 3$ has no integer solutions, because squares are 0 or 1 modulo 4 and no two of those sum to 3. One line kills the whole problem.
3. **Compare sizes of factors.** If $ab = p$ with p prime, then $\{a, b\} = \{1, p\}$ up to sign, and there is nothing left to decide.

STARS AND BARS, AND WHETHER TO CARRY IT

The number of solutions of $x_1 + \cdots + x_k = n$ in **non-negative** integers is $\binom{n+k-1}{k-1}$: picture n stars in a row and $k-1$ bars dropped between them to make k groups, and every arrangement is one solution.

For **positive** integers instead, put one in each box first and then distribute the remaining $n-k$, which gives $\binom{n-1}{k-1}$.

Tier: EDGE. Not named on the specification, it appears occasionally in disguise ("how many ways can 10 identical sweets be shared between 3 children"), and systematic listing is genuinely viable at the sizes this paper uses. Carry the picture, not the formula.

DRILL A RECOGNITION ONLY. NAME THE FIRST MOVE, TEN SECONDS EACH.

1. $xy + 3x + y = 21$ in positive integers.
2. $x^2 - y^2 = 35$ in integers.
3. $x^2 + y^2 = 4k + 3$ in integers.
4. Non-negative integer solutions of $a + b + c = 8$.

ANSWERS. 1. Add 3 to both sides: $(x+1)(y+3) = 24$, then list the factor pairs with $x+1 \ge 2$ and $y+3 \ge 4$. 2. Factor: $(x-y)(x+y) = 35$, then list the factor pairs, negatives included. 3. Squares are 0 or 1 modulo 4, so a sum of two is never 3 mod 4: no solutions. 4. Stars and bars, $\binom{10}{2} = 45$.

DRILL B EXAM STANDARD.

1. Find all positive integer solutions of $xy + 2x + 3y = 18$.
2. Find all integer solutions of $x^2 - y^2 = 21$.
3. Show that $x^2 + y^2 = 2023$ has no integer solutions.
4. How many ways can 10 identical sweets be given to 3 children, if a child may get none?

ANSWERS. 1. Add 6 to both sides: $(x+3)(y+2) = 24$. Positive x, y force $x+3 \ge 4$ and $y+2 \ge 3$, so the admissible factor pairs of 24 are 4×6, 6×4 and 8×3, giving $(x, y) = (1, 4),\ (3, 2),\ (5, 1)$. The pairs 12×2 and 24×1 fail the $y+2 \ge 3$ condition. 2. $(x-y)(x+y) = 21$; both factors have the same parity and 21 is odd, so both are odd, which is consistent. Pairs $(1, 21), (3, 7), (7, 3), (21, 1)$ and their negatives give $x = \pm 11, y = \pm 10$ and $x = \pm 5, y = \pm 2$, eight solutions in all. 3. $2023 = 4 \cdot 505 + 3$, and squares are 0 or 1 mod 4, so a sum of two squares is 0, 1 or 2 mod 4, never 3. 4. $\binom{12}{2} = 66$.

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PART XIV

PAPER 2, PROPERLY

Half the exam, taught almost nowhere, and the cheapest marks in the whole document. The content base is one page long and it can be learned properly in two days.

67 The vocabulary, as a machine

68 Negation and quantifier order

69 Counterexamples, and the two jobs they must do

70 The flawed-step catalogue

71 Proof: choosing the method, ordering the argument

72 "Exactly one of these is true"

Work §67 until the table is automatic. Then §70, which is the part nobody has ever catalogued.

67 The vocabulary, as a machine

The specification's list for Paper 2 is short: true and false, *and*, *or*, *not*, the four if-then forms, converse, contrapositive, necessary, sufficient, for all, for some, there exists, negation, four proof types, and identifying errors in an argument.

That is the entire content base. Most candidates never learn it properly, which is why this is the best return per hour available anywhere in either book.

67.1 Eleven ways to say one thing

67.1 LEARN IT AS A TRANSLATION, NOT AS A MEANING

CORE

TRIGGER Any question about implication, necessity, sufficiency, or which of several statements follow from which.

CLAIM All of the following say **exactly** the same thing, and the set picture is the same for all of them.

WHY An implication is an inclusion of truth sets. $P \Rightarrow Q$ says: every object that makes P true also makes Q true, that is, $P \subseteq Q$.

COST It makes a whole class of question mechanical. Two or three questions a Paper 2, every sitting.

BREAKS Two conditions can be **incomparable**, in which case neither implies the other and the answer is "neither". Do not force an ordering that is not there.

DEFAULT Construct an object satisfying P and check whether it satisfies Q, then the reverse.

ENGLISH	MEANS	SETS
P **implies** Q	$P \Rightarrow Q$	$P \subseteq Q$
If P**, then** Q	$P \Rightarrow Q$	$P \subseteq Q$
Q **if** P	$P \Rightarrow Q$	$P \subseteq Q$
P **only if** Q	$P \Rightarrow Q$	$P \subseteq Q$
P **is sufficient for** Q	$P \Rightarrow Q$	$P \subseteq Q$
Q **is necessary for** P	$P \Rightarrow Q$	$P \subseteq Q$
Q **is an inevitable consequence of** P	$P \Rightarrow Q$	$P \subseteq Q$
P **and not** Q **cannot both hold**	$P \Rightarrow Q$	$P \subseteq Q$
Not Q **implies not** P	$P \Rightarrow Q$	$P \subseteq Q$
P **if and only if** Q	$P \Leftrightarrow Q$	$P = Q$
P **unless** Q	not $Q \Rightarrow P$	complement of $Q \subseteq P$

Nine of those eleven rows are the same statement. That is the point of the table, and it is why the instruction *not to reason about the meaning* is correct: identify the row, draw the inclusion, answer.

THE ONE QUESTION THAT SETTLES MOST OF PAPER 2

Which condition is stricter?

The stricter condition is the smaller set, and the smaller set implies the larger one. Never the other way.

$x > 2$ is stricter than $x > 1$, so $x > 2 \Rightarrow x > 1$. "Is a multiple of 10" is stricter than "is even", so it implies it. "Is a square" is stricter than "is a rectangle". Adding a condition always shrinks the set, so a conjunction implies each of its parts, and each part implies the disjunction.

67.2 "Only if" is the one that costs marks

It is worth slowing down on this row, because it is the one students consistently read backwards.

"P only if Q" does **not** mean "Q implies P". It means P can happen *only* in circumstances where Q also holds — so if Q fails, P cannot happen, which is $P \Rightarrow Q$.

The everyday version makes it obvious. A child insists he cannot have been playing on the tablet for three hours, because the tablet is not even warm. He has used the contrapositive correctly and without instruction: *played for three hours only if the tablet is warm*, so not warm implies not played. The idea is more natural than the phrasing.

67.3 Converse, contrapositive, and the only equivalence

NAME	FORM	STATUS
The statement	$P \Rightarrow Q$	—
The converse	$Q \Rightarrow P$	**Never** guaranteed. Sometimes true, and then only by accident.
The contrapositive	not $Q \Rightarrow$ not P	**Always** equivalent to the original.
The inverse	not $P \Rightarrow$ not Q	Equivalent to the converse, and so also not guaranteed.

That single asymmetry is examined every year, in both directions: sometimes by asking you to identify the contrapositive, sometimes by offering the converse as a plausible option. Two consequences worth holding.

— **If you are asked to prove $P \Rightarrow Q$ and the hypothesis is awkward, prove the contrapositive instead.** It is the same theorem. This is the standard move for statements about integers, about irrationality, and about products being zero.

— **If a question says "which of the following are equivalent to",** the answer set always contains the contrapositive and never contains the converse, unless something extra makes both directions true.

WHY THE CONTRAPOSITIVE IS EQUIVALENT, IN ONE PARAGRAPH

Suppose $P \Rightarrow Q$ holds and suppose Q is false. If P were true, then Q would have to be true as well, and Q cannot be both. So P is false. That is exactly not $Q \Rightarrow$ not P.

The reverse direction is the same argument applied to the contrapositive, using the fact that not(not P) is P.

> You do not need to reproduce that in the exam. You need to have read it once, so that the equivalence is a fact you understand rather than a rule you are trusting.

67.4 The one row of the truth table that matters

$P \Rightarrow Q$ is false in exactly one situation: P **true and** Q **false.** Every other combination leaves it standing.

P	Q	What this case tells you about $P \Rightarrow Q$
true	true	Consistent with it. Does not prove it.
true	false	**Definitely false.** This is the only way to break an implication.
false	true	Consistent with it. Proves nothing either way.
false	false	Consistent with it. Proves nothing either way.

Two things fall straight out of that table, and both are examined.

— **A false hypothesis makes the implication true.** "If $2 + 2 = 5$ then I am the Queen" is a true statement. Nothing follows from a false premise being present, so "if p then q" can only be broken by a case where p actually holds. This is **§69**.

— **A claim about all members of an empty collection is true.** "Every root of $x^2 + 1 = 0$ is positive" is true over the reals, because there are none. **Vacuous truth** appears about once every three papers and it is always the option people cross out first.

And one piece of vocabulary that gets misread: **"some" means at least one, and it does not mean "not all".** "Some primes are even" is true. So is "all even primes equal 2".

DRILL A RECOGNITION. REWRITE EACH AS AN IMPLICATION, TEN SECONDS EACH.

1. n is a multiple of 6 only if n is even.
2. Being differentiable is sufficient for being continuous.
3. $x > 0$ is necessary for $\log x$ to be defined.
4. f is not constant unless f' vanishes somewhere.

ANSWERS. 1. n multiple of $6 \Rightarrow n$ even. 2. differentiable $\Rightarrow$ continuous. 3. $\log x$ defined $\Rightarrow x > 0$. 4. not(f' vanishes somewhere) $\Rightarrow f$ is not constant — equivalently, f constant $\Rightarrow f'$ vanishes somewhere.

DRILL B EXAM STANDARD.

1. Which of these follow from "every multiple of 12 is a multiple of 4"? **(a)** every multiple of 4 is a multiple of 12; **(b)** a number that is not a multiple of 4 is not a multiple of 12; **©** being a multiple of 4 is necessary for being a multiple of 12.
2. A is "$x > 3$" and B is "$x > 1$". Which implies which? Which is necessary for which?
3. "n is prime only if n has exactly two positive divisors." Is it true, and is its converse true?
4. Are "n is even" and "n is a multiple of 3" comparable?

ANSWERS. 1. (b) and ©. Not (a): that is the converse, and 4 is a counterexample. 2. $A \Rightarrow B$, because A is the stricter condition and therefore the smaller set. B is necessary for A; A is sufficient for B. 3. True, and for $n \geq 2$ the converse is true as well, so it is an equivalence. 4. No. Neither implies the other: 4 is even and not a multiple of 3, and 9 is a multiple of 3 and not even. Do not force an ordering that is not there.

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68 Negation and quantifier order

68.1 Mechanical negation

68.1 FLIP EACH QUANTIFIER IN PLACE, THEN NEGATE THE CLAIM

CORE

TRIGGER "Which of the following is the negation of", usually with two or three quantifiers stacked.

CLAIM Split the sentence into its quantified pieces and flip each one **in place, left to right**. "For all" becomes "there exists", "there exists" becomes "for all", and then negate the final claim. Do not paraphrase and do not think about what it means until you have finished.

WHY Negation passes through quantifiers one at a time and the **order is preserved**. That is the whole algorithm.

COST At least one question a sitting is exactly this, and done mechanically it takes twenty seconds with no thought at all.

BREAKS The innermost claim has to be negated too, and it is the step people forget. Also watch *and* and *or*: the negation of "P and Q" is "not P **or** not Q", and the negation of "P or Q" is "not P **and** not Q".

DEFAULT Test the original and each candidate negation on a small example where the original is false. The true negation must be true there.

THE FOUR RULES, WHICH ARE REALLY ONE RULE

$$\neg(\forall x,\ P) \equiv \exists x,\ \neg P \qquad \neg(\exists x,\ P) \equiv \forall x,\ \neg P$$

$$\neg(P \text{ and } Q) \equiv \neg P \text{ or } \neg Q \qquad \neg(P \text{ or } Q) \equiv \neg P \text{ and } \neg Q$$

The bottom row is De Morgan's law, and the top row is the same law with infinitely many statements: "for all" is a long *and*, "there exists" is a long *or*. So there is one instruction: **swap the connective and negate everything inside.**

And one that is not a quantifier rule but is negated wrongly more often than any of them: the negation of $P \Rightarrow Q$ is P **and not** Q. Not "$P \Rightarrow$ not Q". Not "not $P \Rightarrow$ not Q". The negation of an implication is not an implication at all.

68.2 Getting the inner negation right

The negations that are actually wrong on option lists are almost always the innermost claim, not the quantifiers.

$x > 3$	negates to $x \leq 3$, **not** $x < 3$
$x \neq 5$	negates to $x = 5$
n is even	negates to n is odd, for an integer n
p is prime	negates to p is not prime, which for $p \geq 2$ means composite
$f(x) > 0$	negates to $f(x) \leq 0$
at least one	negates to none
exactly one	negates to none **or** more than one
at most two	negates to at least three

68.3 Order of quantifiers changes the claim

68.3 WHETHER THE INNER OBJECT MAY DEPEND ON THE OUTER ONE CORE

TRIGGER	Two different quantifiers in the same sentence, especially "for every … there is …".
CLAIM	"For every n there is an m with $m > n$" is **true**. "There is an m such that for every n, $m > n$" is **false**. Swapping a "for all" and a "there exists" produces a genuinely different statement, and the second form is much stronger, because the object has to work for everything at once.
WHY	In the first form m is allowed to depend on n. In the second it is not. Whether the inner object may depend on the outer one is the entire difference.
COST	This is the standard hard Paper 2 question and it is a free mark once you know to look for it.
BREAKS	Two quantifiers of the **same** type commute freely. "For all x, for all y" is symmetric, and so is "there exist x and y". The trap lives only in the mixed case.
DEFAULT	Read it as a game. The outer quantifier moves first, and "there exists" means you get to choose *after* seeing the previous choices.

A WORKED NEGATION, THREE LAYERS DEEP

Take a definition of the kind Paper 2 likes to invent:

*"f is **positively increasing** if and only if, for every real a, there exists $h > 0$ such that, for every real x with $a - h \le x \le a + h$, if $f(x) > 0$ then $f'(x) > 0$."*

Negate it mechanically, one layer at a time, without thinking about what it means.

1. "For every real a" becomes "there exists a real a".
2. "There exists $h > 0$" becomes "for every $h > 0$".
3. "For every real x with $a - h \le x \le a + h$" becomes "there exists a real x with $a - h \le x \le a + h$". Note the **range condition stays put** — it is part of the description of x, not part of the claim.
4. The innermost claim is an implication, $f(x) > 0 \Rightarrow f'(x) > 0$, whose negation is $f(x) > 0$ **and** $f'(x) \le 0$.

So: *there exists a such that for every $h > 0$ there exists x with $a - h \le x \le a + h$ satisfying $f(x) > 0$ and $f'(x) \le 0$.*

Four mechanical steps, no interpretation, twenty seconds. Now look at what the distractors on such a question are made of: one leaves a quantifier unflipped, one negates the range condition instead of the claim, one turns the inner implication into another implication, one flips the quantifiers but in the wrong order. Every one of those is a step you can name.

DRILL A RECOGNITION ONLY. NEGATE, TEN SECONDS EACH.

1. $x \ge 5$.
2. "P and not Q".
3. "For every n there exists m such that $P(n, m)$".
4. "p implies q".

ANSWERS. 1. $x < 5$. 2. "Not P or Q". 3. "There exists n such that for every m, not $P(n, m)$" — flip both, keep the order. 4. "p **and** not q". The negation of an implication is not an implication.

DRILL B EXAM STANDARD.

1. Negate: "For every prime p, p is odd."
2. Negate: "There exists an integer n such that $n^2 < n$."
3. Negate: "For every positive integer N there is a positive integer K such that for every positive integer m, $N(Km+1)-1$ is not prime."
4. Negate: "$x > 3$ and $y \leq 2$."

ANSWERS. 1. There exists a prime p that is not odd. (True: $p = 2$.) 2. For every integer n, $n^2 \geq n$. (True.) 3. There exists a positive integer N such that for every positive integer K there is a positive integer m for which $N(Km+1)-1$ is prime. 4. $x \leq 3$ or $y > 2$.

69 Counterexamples, and the two jobs they must do

69.1 THE HYPOTHESIS FIRST, THE CONCLUSION SECOND

CORE

TRIGGER	"Which of these is a counterexample to", or a question asking you to disprove a claim.
CLAIM	To break "if p then q" you need one case where p is **true** and q is **false**. An object that fails p proves nothing at all, however badly it fails q.
WHY	The claim only ever asserts something about objects satisfying p. Anything outside that set is simply not being talked about.
COST	An enormous number of marks are lost on the first job rather than the second, and the exam builds options out of exactly that error.
BREAKS	Nothing, but check both jobs out loud, every time.
DEFAULT	Check every option against both conditions. On five options that takes ninety seconds and it is never wrong.

WHERE TO LOOK FOR THE BREAKING CASE, IN THIS ORDER

1. **The boundary.** If the claim mentions $x > 2$, look at x just above 2. If it mentions positive integers, look at 1.
2. **Zero, one, and the empty case.** Most false claims about products break at zero and most false claims about ratios break at one.
3. **Negative numbers, then non-integers.** Claims stated for "numbers" are often only true for positive integers, and the author forgot to say so.
4. **Equality inside an inequality**, and repeated elements inside a claim about distinctness.
5. **The degenerate configuration** in geometry: three collinear points, a zero-radius circle, a tangent instead of a secant.

69.2 Counterexamples to nested statements

The harder version, and the one that separates candidates. When the claim has a quantifier inside it, breaking the outer statement is not the same as breaking one instance of the inner one.

THREE EXAMPLES, IN INCREASING DIFFICULTY

Simple. *"If n can be written as a sum of two squares, then n is not a cube."* Take $n = 8 = 2^2 + 2^2$, which satisfies the hypothesis, and $8 = 2^3$, so the conclusion fails. Valid.

Now the invalid one that looks fine: $n = 27 = 3^3$ is a cube, but 27 is not a sum of two squares (the squares up to 27 are $0, 1, 4, 9, 16, 25$ and no two sum to 27), so it fails the hypothesis and proves nothing.

With an inner "there exists". *"If $n > 3$ is a positive integer, then there exists a positive integer $m < n$ such that every integer r with $m < r < m + 3$ is a factor of n."*

Is $n = 6$ a counterexample? Trying $m = 2$ gives $r = 3, 4$, and 4 is not a factor of 6, so that m fails. But the claim only asks for **some** m, and $m = 1$ gives $r = 2, 3$, both factors of 6. So $n = 6$ is not a counterexample. To break it you must show **every** m fails: $n = 9$ works, because the factors of 9 are $1, 3, 9$ and no two of them differ by 1, so no m can succeed.

With an inner implication. *"If n is a multiple of 4 greater than 4, then for every integer m with $2 < m < n$, if m is a factor of n then m is even."*

$n = 12$ with $m = 3$ works: 12 satisfies the outer hypothesis, $2 < 3 < 12$, 3 is a factor, and 3 is not even. Both inner jobs done, both outer jobs done.

$n = 10$ with $m = 5$ fails, because 10 is not a multiple of 4. $n = 12$ with $m = 5$ fails, because 5 is not a factor of 12, so the inner implication is never engaged.

The pattern to extract: **to break a "for all" you need one case; to break a "there exists" you need all cases.** Which one you are facing is decided by the quantifier nearest the thing you are attacking, and reading that correctly is the whole skill.

DRILL A RECOGNITION ONLY. TEN SECONDS EACH.

1. To break "if p then q", what two things must your example do?
2. A claim about all $x > 2$. Where do you look first?
3. "If n is a multiple of 4 then n is a multiple of 8." Give one counterexample.
4. Somebody offers $n = 7$ against "if n is even then n^2 is even". Verdict?

ANSWERS. 1. Satisfy p, and fail q. Both, every time. 2. Just above the boundary: x a little more than 2. 3. $n = 12$. 4. It proves nothing: 7 fails the hypothesis, so the claim never spoke about it.

DRILL B EXAM STANDARD.

1. Which is a counterexample to "if n is odd then $n^2 + n$ is odd"? $n = 3$, $n = 4$, or neither?
2. Give a counterexample to "if $x^2 > 4$ then $x > 2$".
3. Give a counterexample to "if a quadrilateral has four equal sides then it is a square".
4. Give a counterexample to "if $a < b$ then $a^2 < b^2$".

ANSWERS. 1. $n = 3$ gives $n^2 + n = 12$, even, so the hypothesis holds and the conclusion fails: it is a counterexample. $n = 4$ fails the hypothesis and proves nothing. 2. $x = -3$. 3. A rhombus with angles $60°, 120°, 60°, 120°$. 4. $a = -3$, $b = 1$.

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70 The flawed-step catalogue

Error-spotting questions are on the specification by name, and the flawed step is almost always one of these. Learn the list and the question becomes a lookup.

THE FLAW	WHAT IT LOOKS LIKE
Dividing by something that could be zero	$ab = ac$, therefore $b = c$. Lost the case $a = 0$.
Squaring without checking signs	Turns $\sqrt{x} = x - 2$ into $x = x^2 - 4x + 4$ and invents a root.
Keeping only the positive square root	$x^2 = 9$ has two solutions. And $\sqrt{x^2} = \lvert x \rvert$, not x.
Concluding $A = B$ from $\sin A = \sin B$	Or from $\cos A = \cos B$, or $\tan A = \tan B$. All are many-to-one.
Multiplying an inequality by a variable of unknown sign	§48.1. The most common one on this paper.
Using the converse	Proving B from A and then claiming A from B.
Assuming the conclusion	Usually disguised as an obvious step somewhere in the middle. Watch for a line that is what you were trying to prove.
Checking finitely many cases	Then concluding a statement about all integers. $n^2 + n + 41$ is prime for $n = 0$ to 39 and not for $n = 40$.
Applying a formula outside its conditions	A geometric sum to infinity with $\lvert r \rvert \geq 1$; a logarithm of something that could be non-positive; a discriminant on something that is not a quadratic.
Losing a case in a modulus	Solving $\lvert x - 1 \rvert = 3$ and reporting one answer.
Treating a necessary condition as sufficient	Most commonly: checking that a stationary point exists and concluding it is a maximum.
Reversing a step that is not reversible	Squaring, multiplying by zero, applying a many-to-one function. Each of these is a one-way implication and the argument needs it both ways.

HOW TO ACTUALLY USE THAT LIST

Not by reading it. Take three purported proofs from official Paper 2 papers, find the flaw, then **name which row of the catalogue it was.**

After ten of those the lookup is automatic, and you will start to notice the same errors in your own working, which is worth more on Paper 1 than the Paper 2 marks were.

70.1 The implication arrow as a habit

Most flawed steps are one symbol wrong. When you write a chain of working, each line follows from the one before by either $\Rightarrow$ or $\Leftrightarrow$, and knowing which is not pedantry.

Take $x = \sqrt{x + 2}$. Squaring gives $x^2 = x + 2$, so $x = 2$ or $x = -1$. But squaring is only $\Rightarrow$, not $\Leftrightarrow$, so those are *candidates*, not solutions. Checking: $x = 2$ works and $x = -1$ does not, because $\sqrt{1} = 1 \neq -1$.

> *If every step in your chain is $\Leftrightarrow$, you are finished. If any step is only $\Rightarrow$, you must check your answers in the original.*

That single habit prevents a large fraction of the errors in the catalogue above, and it is the reason the specification asks about it.

71 Proof: choosing the method, ordering the argument

Four proof methods are named. Choosing between them takes five seconds and saves four minutes.

METHOD	USE IT WHEN	THE SHAPE
Direct	The hypothesis is easy to work with.	Assume P, deduce Q in a chain.
Contrapositive	The hypothesis is awkward but its negation is concrete.	Assume not Q, deduce not P. Standard for "if n^2 is even then n is even".
Contradiction	The claim is a non-existence or an irrationality, or nothing else has worked.	Assume the opposite, derive something impossible, conclude.
Exhaustion	The claim is about a finite, small collection, or splits into a few cases.	Check every case. Say explicitly that you have covered them all.

> **THE TWO CLASSICS, COMPRESSED TO THEIR SPINE**
>
> **Infinitely many primes.** Suppose there were finitely many, $2, 3, 5, \ldots, P_n$. Form $Q = 2 \cdot 3 \cdot 5 \cdots P_n + 1$. Q leaves remainder 1 on division by every prime in the list, so no prime in the list divides it; but every integer above 1 has a prime factor, so Q has one outside the list. Contradiction.
>
> **$\sqrt{2}$ is irrational.** Suppose $\sqrt{2} = a/b$ in lowest terms. Then $a^2 = 2b^2$, so a^2 is even, so a is even (contrapositive of "odd squared is odd"), so $a = 2k$ and $4k^2 = 2b^2$, so $b^2 = 2k^2$ and b is even too. Both even contradicts lowest terms.
>
> Neither is examinable as a piece of writing. Both are examinable as a **shape**: assume the opposite, derive something impossible, conclude. And both contain the move that is examinable in its own right — "a^2 even implies a even" is proved by contrapositive, not directly.

71.1 Choosing the case split

Proof by cases is on the specification and the only skill in it is choosing the split so that each case becomes trivial. The options, in order of how often they work:

— **Parity**, so even and odd;

— **Sign**, so positive, negative and zero;

— **Residue modulo 3 or 4**, for anything with squares in it;

— **Whether a particular factor is zero.**

If your case split leaves you with the same difficulty in both branches, you chose the wrong split.

71.2 Rearranging a proof into order

71.2 A PROOF IS A DEPENDENCY ORDER — CORE

TRIGGER A list of statements, asked for the order that makes a valid proof.

CLAIM	Work from both ends. The first line is the one that depends on nothing, usually the hypothesis or a definition. The last line states the target. Then chain the middle by asking, for each line, **which object or fact it needs to already exist**. A line that introduces a new symbol must come after the line that defines it.
WHY	A proof is a dependency order, so the question is a sorting problem with a rule, not a mathematical problem.
COST	Two minutes of reading and rereading becomes forty seconds.
BREAKS	Two lines can be genuinely interchangeable, so more than one order can be valid. Check what the options actually differ on before deciding.
DEFAULT	Take each offered order and read it as a proof. The invalid ones usually break within two lines.

71.3 Conjecture from small cases

71.3 THREE CASES, THEN AN AWKWARD FOURTH CORE

TRIGGER	A pattern, a sequence of expressions, or "make a conjecture and justify it".
CLAIM	Compute three cases. Guess the form. Test a fourth, **chosen to be awkward rather than convenient**. Then check a boundary case. Only then commit.
WHY	Three cases identify a linear or quadratic pattern and almost nothing else, so the fourth is where a wrong guess dies. The specification asks for exactly this: conjecture from small cases, then justify.
COST	It is the intended route for a whole class of question, and it is also the level 2 rung of the ladder in Book One §9, which is why that ladder is worth more on Paper 2 than on Paper 1.
BREAKS	Convenient fourth cases confirm wrong guesses. If your first three cases were $n = 1, 2, 3$, choose $n = 10$ or $n = 0$, not $n = 4$.
DEFAULT	Algebra from the start, which is slower and often impossible before you know what you are proving.

DRILL A RECOGNITION ONLY. NAME THE METHOD, TEN SECONDS EACH.

1. "If n^2 is odd then n is odd."
2. "$\sqrt{3}$ is irrational."
3. "Every integer from 1 to 5 satisfies P."
4. A claim about n^2 for all integers n. Which case split?

ANSWERS. 1. Contrapositive: assume n is even and square it. 2. Contradiction. 3. Exhaustion, and say explicitly that you have covered every case. 4. Parity, or residues modulo 3 or 4 — whichever makes the target trivial.

DRILL B EXAM STANDARD.

1. Is this a valid argument? "$x^2 > x$, so dividing by x gives $x > 1$."
2. "For all real x, if $x > 2$ then $x^2 > 4$." State the converse and the contrapositive, and say which is true.
3. Prove that if n^2 is divisible by 3 then n is divisible by 3.
4. Find the flaw: "Let $a = b$. Then $a^2 = ab$, so $a^2 - b^2 = ab - b^2$, so $(a - b)(a + b) = b(a - b)$, so $a + b = b$, so $a = 0$."

ANSWERS. 1. No. Dividing by x requires $x > 0$; for $x = -1$ the hypothesis holds and the conclusion fails. 2. Converse: if $x^2 > 4$ then $x > 2$, false at $x = -3$. Contrapositive: if $x^2 \leq 4$ then $x \leq 2$, true, because a contrapositive

is always equivalent to a true statement. 3. Contrapositive: if $3 \nmid n$ then $n = 3k \pm 1$, so $n^2 = 9k^2 \pm 6k + 1$, which is 1 more than a multiple of 3, so $3 \nmid n^2$. 4. Dividing both sides by $a - b$, which is zero.

72 "Exactly one of these is true"

A whole archetype, worth its own section because the method is completely mechanical and almost nobody is taught it.

The question gives six or eight statements and tells you that exactly one of them is true. You are not asked to evaluate any of them on its own. You are asked to use the *structure of the list*.

THE THREE-STEP METHOD

1. **Rewrite every statement in one normal form.** Choose a direction — "condition on the variable ⇒ condition on the function" is usually the right one — and put every option into it, using the contrapositive wherever needed.
2. **Find the duplicates.** If two statements turn out to be the same statement, neither can be the unique true one, so **both are eliminated**. This usually kills half the list in twenty seconds.
3. **Find the implications.** If statement X implies statement Y, then X cannot be the unique true one, because X being true would make Y true as well. Eliminate X. What survives is the answer.

WORKED, WITH THE ELIMINATION SHOWN

"Exactly one of the following is true for all real x, for a real-valued function f and a real number a. Which is it?"

1. $f(x) \leq 0$ if $x > a$.
2. $x \leq a$ is equivalent to $f(x) > 0$.
3. $f(x) > 0$ only if $x \leq a$.
4. $x > a$ and $f(x) > 0$ cannot both be true.
5. $f(x) \leq 0$ is a necessary condition for $x > a$.
6. $f(x) > 0$ is a sufficient condition for $x \leq a$.
7. $f(x) < 0$ and $x \leq a$ cannot both be true.
8. $x > a$ implies $f(x) \leq 0$.

Normalise every one into "condition on x ⇒ condition on f", using §67.1:

— 1, 3, 4, 5, 6 and 8 all become $x > a \Rightarrow f(x) \leq 0$.

— 2 becomes $x > a \Leftrightarrow f(x) \leq 0$.

— 7 becomes $x \leq a \Rightarrow f(x) \geq 0$.

Now eliminate. Statements 1, 3, 4, 5, 6 and 8 are identical, so if any of them were true, six would be true; since exactly one is, that whole block is false. Statement 2 is an equivalence, so it *implies* statement 1; if 2 were true then 1 would be too, so 2 is out.

Only 7 remains, so the answer is **7** — and notice that you never had to decide whether any individual statement was true. The structure did all of it.

THE TWO WAYS THIS GOES WRONG

Not normalising. If you leave the options in their English forms you will compare "only if" with "necessary for" and get lost. Rewrite all of them first, in the same direction, before you look at any of them.

Forgetting that an equivalence implies both halves. A biconditional is strictly stronger than either implication in it, so it can never be the unique true statement when one of its halves is also on the list. That is the elimination people miss.

DRILL A RECOGNITION ONLY. REWRITE AS "CONDITION ON X ⇒ CONDITION ON P", TEN SECONDS EACH.

1. P only if $x < 1$.
2. $x \geq 2$ is sufficient for P.
3. P is necessary for $x < 0$.
4. P and $x > 5$ cannot both be true.

ANSWERS. 1. $P \Rightarrow x < 1$, so in the required direction $x \geq 1 \Rightarrow \neg P$. 2. $x \geq 2 \Rightarrow P$. 3. $x < 0 \Rightarrow P$. 4. $x > 5 \Rightarrow \neg P$.

DRILL C THE 9.0 LAYER.

1. Exactly one of the following is true about a statement P depending on x. Which? **(A)** If $|x| < 1$ then P. **(B)** P false and $|x| \leq 1$ cannot both hold. **(C)** $0.5 < x < 2.5$ is necessary for P. **(D)** $|x| < 2$ only if P. **(E)** $0.5 < x < 2.5$ if P. **(F)** P is false if and only if $|x| \geq 1$.

ANSWERS. Normalise to "condition on $x \Rightarrow P$ or $\neg P$". (A) $|x| < 1 \Rightarrow P$. (B) $|x| \leq 1 \Rightarrow P$. (C) $x \leq 0.5$ or $x \geq 2.5 \Rightarrow \neg P$. (D) $|x| < 2 \Rightarrow P$. (E) $x \leq 0.5$ or $x \geq 2.5 \Rightarrow \neg P$. (F) $|x| \geq 1 \Leftrightarrow \neg P$. Now: C and E are identical, so both go. D implies B, because $|x| \leq 1$ sits inside $|x| < 2$; so D goes. B implies A, because $|x| < 1$ sits inside $|x| \leq 1$; so B goes. F is an equivalence containing $|x| < 1 \Rightarrow P$, so F implies A and F goes. Only A can stand alone.

15

PART XV

THE 9.0 LAYER

For the reader who found the rest easy. Pricing your own shortcuts, using structure to kill options before computing anything, and questions that need two chapters at once.

73 Ratio, invariance and scaling

74 The break-even rule

75 Invariance as an eliminator

76 Composite drills

This part assumes Parts X to XIV are automatic. If they are not, it will just feel like showing off.

73 Ratio, invariance and scaling

The first printing had four separate ratio results. They are all the same observation, and the observation is worth more than the four results because it applies to configurations nobody has tabulated.

73.1 The scaling law behind every ratio result

73.1 A RATIO OF TWO AREAS IS A PURE NUMBER

CORE

TRIGGER A question asking for a ratio of areas, or for a value that makes two areas equal, with the constants left general or deliberately grotesque.

CLAIM In any of the configurations of §54 the enclosed area is a fixed number times $|a|$ times a width to the power $\deg + 1$. So a ratio of two such areas **cannot depend** on a, on where the configuration sits, on the tilt of any line, or on the vertical scale. Concretely: for $y = ax^n$ with $a, k > 0$, the curve cuts the rectangle with corners $(0, 0)$ and (k, ak^n) into two pieces in the ratio n above the curve to 1 below it, for every $n \geq 1$. So a parabola cuts it $2 : 1$, a cubic $3 : 1$, a quartic $4 : 1$.

WHY $\int_0^k ax^n \, dx = \dfrac{ak^{n+1}}{n+1}$ while the whole rectangle is ak^{n+1}. So the piece below the curve is $\frac{1}{n+1}$ of the rectangle and the piece above it is $\frac{n}{n+1}$. Everything cancels except n, which is the entire reason these ratios are constants.

COST On a question asking "find q so that the two regions are equal", this is a five-second tick against a page of integration.

BREAKS The corner has to be where the curve actually is, that is at the origin for $y = ax^n$ and at the stationary point or centre after any shift. If the rectangle is drawn from an arbitrary point, the ratio is not $n : 1$. And for a curve that is not a single power, there is no such constant: check the form before you use it.

DEFAULT Integrate and divide.

73.2 The sum of intersections is fixed

73.2 A LINE CANNOT REACH THE TOP TWO COEFFICIENTS

EDGE

TRIGGER Two different lines drawn across the same cubic, with some intersections given and one missing. Often one of the lines is tangent somewhere.

CLAIM For a fixed cubic $ax^3 + bx^2 + cx + d$, any line meets it where $ax^3 + bx^2 + (c - m)x + (d - k) = 0$. A line changes only the last two coefficients, so the **sum of the three intersection x-coordinates is always $-b/a$**, whatever line you draw. Count a tangency twice.

WHY The sum of the roots is $-b/a$ (§45.1) and the line cannot reach b or a.

COST It converts "find the third intersection" into a subtraction. Ten seconds, and it appears more often than its obscurity suggests.

BREAKS All three roots must be real for you to be reading them off a picture. If a line meets the cubic once, the other two roots exist but are not on your diagram, and the sum still holds while being useless. Also, **a repeated root counts twice**: this is where people lose the mark.

DEFAULT Set up and solve the cubic.

73.3 Where a cubic's key points sit

73.3 THE FIVE POINTS ARE EQUALLY SPACED EDGE

TRIGGER	A cubic with its turning points, its centre, and horizontal lines through the turning points all in play.
CLAIM	Take the turning points at $x = \pm h$ from the centre. Then the horizontal line through each turning point meets the curve again at $\mp 2h$, so the five key x-values are $-2h, -h, 0, h, 2h$, equally spaced. The horizontal line through the centre meets the curve at $\pm\sqrt{3}\,h$.
WHY	Every cubic is $a(x^3 - 3h^2x)$ after a shift, because the shift kills the quadratic term and the scale fixes the rest. Then $x^3 - 3h^2x = 2h^3$ factorises as $(x + h)^2(x - 2h)$, and $x^3 - 3h^2x = 0$ gives $0, \pm\sqrt{3}h$. Two factorisations, thirty seconds.
COST	Turns a ratio question into reading off a number line.
BREAKS	Points of inflexion are explicitly not examined beyond the qualitative, so this is EDGE at best and appears mostly in disguise, as a question about two horizontal lines and the distances between their intersections.
DEFAULT	Differentiate, find the turning points, substitute, solve.

73.4 Ratios as equations

A smaller point, and one that costs marks in applied-context questions, which are now more common.

- **A ratio is not a fraction until you name the total.** $a : b = 3 : 5$ means $a = 3k$ and $b = 5k$ for some k, and introducing that k immediately is almost always the right first move. It converts a ratio into two equations.
- **Ratios compose by matching the shared term.** $a : b = 2 : 3$ and $b : c = 4 : 5$ give $a : b : c = 8 : 12 : 15$, by scaling each ratio so the b entries agree.
- **A ratio of parts is not a ratio to the whole.** If $a : b = 3 : 5$ then a is $\frac{3}{8}$ of the total, not $\frac{3}{5}$. Two of the five options will be built from exactly that confusion.

74 The break-even rule

Every shortcut in this book, and every shortcut in anything else you will ever be sold, is an investment with a return you can estimate. The objection that "applying the formula takes longer than the original method" is not a complaint about formulae. It is an observation that most people never do this arithmetic.

Write f for how often the cue appears, t_d for the default time, t_s for the shortcut time **including checking the conditions**, and p for the chance you fire it wrongly. Then the shortcut is worth holding when

$$f\,(t_d - t_s) \;>\; p \times (\text{cost of a wrong answer}) \;+\; \text{recall cost}.$$

You do not need numbers. You need the four questions, and asking them takes ten seconds per result.

RESULT	FREQUENCY	SAVES	VERDICT
$\lvert a\rvert D^3/6$, the quadratic area	Every paper	3 min	**Learn cold.** The clearest win in the document.

RESULT	FREQUENCY	SAVES	VERDICT
Discriminant conditions	Every paper	Route	Not a shortcut, the method.
Sum and product of roots	Most papers	90 s	**Learn cold.**
The substitution constraint table (§46)	Most papers	Marks	**Learn cold.** It is not a shortcut, it is the half of the method people omit.
Divisor counting	Most papers	60 s	**Learn cold.**
Mechanical negation	Every Paper 2	2 min	**Learn cold.**
$\lvert a\rvert D^4/12$**, tangent cubic**	Every few papers	3 min	Learn, and check the tangency algebraically.
Two-lobe cubic areas	Rare	2 min	Learn the **splitting trick**, not the formula.
Telescoping sums	Rare	3 min	Learn the picture. The formula is one line from it.
Power of a point	Rare	60 s	Know it, prefer the similar triangles.
Quartic area constants	Rare	3 min	Know the identity, look up the constant.
Stars and bars	Rare	90 s	Only if you can also derive it.
15° **exact values**	Never	0	Do not carry it into the exam.
Ceva, Menelaus, Stewart	Never	0	§77.

Two rules fall out of that table and both are worth more than the table.

> **THE FIVE SECOND RULE**
>
> If you cannot confirm a result's conditions within five seconds of seeing the question, **use the default method.**
>
> Being right slowly beats being wrong quickly, and on this exam it beats it by exactly one mark plus your composure for the next question.

> **PREFER IDENTITIES TO FORMULAE**
>
> An identity you can rederive has **no misfire risk**, because deriving it re-checks its conditions on the way.
>
> That is the whole argument for §54 replacing eight memorised results with one, for the splitting trick beating the two-lobe formula it produces, and for §53.3 telling you to derive the segment rather than remember it.

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75 Invariance as an eliminator

Book One §22 lists five ways to use the option list. This is the content-side sixth, and it is the one that separates a fast solver from a very fast one.

> *Before computing anything, ask what the answer cannot possibly depend on.*

On this exam the answer is usually startling.

INVARIANCE	WHAT IT KILLS
Scaling	If the configuration is an area between a degree n polynomial and a line, the answer must be a number times $\lvert a\rvert$ times a width to the power $n+1$. Any option not of that shape is dead without a calculation. If the question doubles a width, the answer multiplies by 2^{n+1}.
Translation	Nothing in a question about separations, ratios or areas can depend on **where** the configuration sits. So an option containing the value of c, or the roots themselves rather than their difference, is dead.
Parity and residues	If the answer is an integer built from an even count, odd options are dead. If the answer is a number of solutions, it is a small non-negative integer, and options that are not can go.
Sign and size	Establish the sign in five seconds and the order of magnitude in ten. On a question with five options this routinely leaves two.
Dimension	An answer that should be a length cannot be an area. In a question with r and θ in it, an option containing $r\theta^2$ is neither.
Symmetry of the roles	If the question treats two objects identically, the answer must be symmetric in them. An option that is not symmetric is dead. This one is under-used and it is very cheap.

75.1 Reading the option list for the intended method

The options leak the route. This is a content skill because what leaks is content.

THE OPTIONS LOOK LIKE	SO THE QUESTION IS ABOUT
$a+b\sqrt{3}$	An exact trig value at 30° or 60°, or an equilateral triangle.
$a+b\sqrt{2}$	45°, or a square, or a nested-shapes ratio.
Multiples of π, or π and a surd together	Radians, and probably a sector or a segment.
Two options differing by a factor of 2	A halving step: a sector against a triangle, a chord against a half-chord, an area against its double.
Two options differing only in sign	A direction, an orientation, or which curve is on top.
Four integers and one that is not	Either the answer or the designed error, and it is worth three seconds to work out which.
Factorials or towering powers left unsimplified	An invariance question. The monstrous constant is meant to cancel, see §54.
A logarithm in every option	Take logarithms of the equation before you start.
Options differing only by $<$ against $\leq$	A boundary case, an endpoint, or a denominator that vanishes (§48).
Options that are counts: 0, 1, 2, 3, 4	Sketch and count. Do not solve (§8.3).

76 Composite drills

Every question below needs at least two sections of this book. Target three minutes each, and record **which cue you missed** rather than whether you got the number.

DRILL C COMPOSITES, THREE MINUTES EACH.

1. The line $y = c$ meets $y = x^3 - 6x^2 + 9x$ at three points whose x-coordinates are in arithmetic progression. Find c, and then the total area between the line and the curve.
2. For how many integers k with $1 \le k \le 50$ does $x^2 + kx + k = 0$ have two distinct real roots?
3. $f(x) = \log_2\!\left(x^2 - 6x + 10\right)$. Find the least value of f and where it occurs.
4. A circle passes through $(0,0)$ and $(6,0)$ and is tangent to the line $y = x$. How many such circles are there, and what is the radius?
5. Prove or disprove: for every positive integer n, $n^2 + n + 41$ is prime.
6. $x_1 = 3$ and $x_{n+1} = 1 - \dfrac{1}{x_n}$. Find x_{100}.
7. The tangents to $y = x^2$ at $x = p$ and $x = q$ meet at a point on the line $y = -1$. Show that $pq = -1$, and find the area between the parabola and the chord joining the two points of tangency when $p = -2$.
8. How many of the divisors of $2^6 \cdot 3^4 \cdot 5^2$ are multiples of 12?
9. $f(x) + f(6 - x) = 8$ for all x, and f is continuous. Find $\displaystyle\int_2^4 f(x)\,dx$.
10. How many solutions does $\left|x^2 - 4\right| = k$ have, in terms of k?

ANSWERS. **1.** The sum of the roots is 6 whatever c is (**§73.2**), so the middle root is 2 and $c = f(2) = 2$. Then $x^3 - 6x^2 + 9x - 2 = (x-2)(x^2 - 4x + 1)$, whose roots are $2 \pm \sqrt{3}$, so the gaps are equal at $d = \sqrt{3}$ and each lobe is $|a|d^4/4 = 9/4$; total $9/2$ (**§54**). **2.** $\Delta = k^2 - 4k > 0$ needs $k < 0$ or $k > 4$, so k from 5 to 50, which is **46** (**§44**, **§64**). **3.** The inside completes the square to $(x-3)^2 + 1$, least value 1 at $x = 3$, and $\log_2 1 = 0$ (**§47**, **§48**). **4.** The centre is on $x = 3$ by the perpendicular bisector, so it is $(3, t)$ with radius $\sqrt{9 + t^2}$; tangency to $y = x$ needs $|3 - t|/\sqrt{2} = \sqrt{9 + t^2}$, so $9 - 6t + t^2 = 18 + 2t^2$, giving $t^2 + 6t + 9 = 0$ and $t = -3$. **One** circle, centre $(3, -3)$, radius $3\sqrt{2}$ (**§57**, **§58**). **5.** False: $n = 40$ gives $41 \cdot 41$. The point is that the first thirty-nine cases all work, which is the finite-checking error from **§70**. **6.** Period 3: $3, \tfrac{2}{3}, -\tfrac{1}{2}, 3, \ldots$; $100 = 3 \cdot 33 + 1$, so $x_{100} = 3$ (**§63**). **7.** The tangents at p and q meet at $\left(\tfrac{p+q}{2},\ pq\right)$, so $pq = -1$; then $q = \tfrac{1}{2}$, $D = \tfrac{5}{2}$, and the area is $\tfrac{1}{6}\left(\tfrac{5}{2}\right)^3 = \tfrac{125}{48}$ (**§54**). **8.** A multiple of 12 needs at least 2^2 and 3^1, so the exponents run 2 to 6, 1 to 4 and 0 to 2, giving $5 \cdot 4 \cdot 3 = 60$ (**§64**). **9.** Centre at $p = 3$ with $c = 8$, and $[2, 4]$ is centred on 3 with $h = 1$, so the integral is 8 (**§51**). **10.** Sketch $y = |x^2 - 4|$, a W-shape with minima at ± 2 and a local maximum of 4 at $x = 0$. Then: no solutions for $k < 0$; two for $k = 0$; four for $0 < k < 4$; three for $k = 4$; two for $k > 4$ (**§50.4**, **§43.4**).

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PART XVI

BEYOND, AND HOUSEKEEPING

The material that is genuinely off the specification and worth keeping anyway, the ESAT differences, an honest account of what was wrong in the previous edition, and where to find everything named in this book.

77 Beyond the specification

78 If you are sitting the ESAT

79 What changed in this edition, and what was corrected

80 Where to find everything

Read §77 when you have time to spare, never in the last month. Read §80 whatever you do: every link in it is live.

77 Beyond the specification

Everything here is BEYOND. None of it can be required by a question, all of it is real mathematics, and most of it pays somewhere else. The reason it is in a separate section rather than deleted is that the somewhere else matters: Oxford and Cambridge interviews, the MAT, STEP, and the first year of a degree.

RESULT	WHERE IT ACTUALLY PAYS
$\sin 15^\circ$, $\cos 15^\circ$, and the $\sqrt{6} \pm \sqrt{2}$ triangle	Anywhere compound angle formulae are allowed, which is the second year of A-level onwards. Derive it from $\sin(45^\circ - 30^\circ)$ rather than memorising the triangle.
Cauchy–Schwarz and AM–GM in full generality	Interviews, olympiad, STEP. The two-variable cases are completed squares, which is why this exam does not need them (§48.3).
Vieta's formulae for higher degree	Any question about symmetric functions of the roots of a quartic or beyond. Genuinely useful at interview.
Trinomial and multinomial general terms	Combinatorics later. The specification stops at $(1 + x)^n$ and $(a + f(x))^n$.
Ceva, Menelaus, Stewart, Apollonius	Olympiad geometry, and one good interview answer about why concurrence is a ratio condition.
Quantitative work with points of inflexion	The second year of A-level and STEP. Explicitly excluded here.
Quartic area constants, $\frac{1}{20}$ and $\frac{1}{30}$	They come free from §54, so there is nothing extra to learn. Do not drill them.
Formal ε–δ definitions of limits and continuity	Nothing on this paper. But the *negation* of such a definition is exactly the nested-quantifier exercise in §68, and working one through is the best possible practice for it.

> **ONE HONEST NOTE ABOUT ALL OF IT**
>
> Studying this material is one of the more enjoyable things you can do with a summer, and it builds the kind of mathematical taste that shows at interview.
>
> It is also, in the eight weeks before a sitting, the single most attractive way to avoid the work that would actually move your score. Book One §29 has the general form of that trap: it is "practising what you are already good at" wearing a more flattering costume.

78 If you are sitting the ESAT

The first printing described itself as covering the TMUA and the ESAT together. They are different tests and the differences change the strategy more than the content.

	TMUA	ESAT
Shape	Two papers of 20 questions in 75 minutes	Modules of 27 questions in 40 minutes, usually three of them, separately timed
Pace	About 3 min 45 s a question	About 89 seconds a question

	TMUA	ESAT
Maths modules	Papers 1 and 2, both compulsory	Mathematics 1 compulsory; Mathematics 2 only for some courses
Reasoning	Paper 2 is half the exam	No logic and proof paper at all
Content	AS pure, plus a GCSE layer	Mathematics 1 is GCSE level; Mathematics 2 adds calculus, radians, logarithms, and the sine and cosine rules
Marking	No penalty for a wrong answer	The same

What follows for a reader using this book for the ESAT.

— **Part XIV does not apply to you for the test itself.** Read it anyway if you have the time, because the flawed-step catalogue in §70 improves your own accuracy on every other question you will ever do.

— **Halve every time budget.** At 89 seconds a question the profile in Book One §1.2 compresses hard: you need many more questions arriving under thirty seconds, which raises the value of everything labelled CORE and lowers the value of everything else.

— **Time does not carry between modules.** Finishing early buys you nothing, so there is no reason to rush the end of a module you can still work on.

— **Sort this book by module.** For Mathematics 1, the load-bearing sections are §48, §50, §57 and §64. For Mathematics 2, add §44, §47, §53 and §54.

79 What changed in this edition, and what was corrected

An edition that does not say what it changed is asking to be trusted without evidence. Here is the account.

79.1 Structural changes

— **Two volumes became one book in two halves,** with a single continuous section numbering, so a cross-reference means the same thing everywhere. Book One's numbering, §1 to §39, is unchanged from the previous edition.

— **The content half was reorganised by mathematical area** rather than by the order results happened to be written in, and re-indexed on what you *see* in a question rather than on the topic a result belongs to. The bridge is the cue table in Book One §12.3, which is new.

— **Seven areas the exam leans on gained sections from scratch:** polynomial structure and the factor theorem (§43), roots and coefficients (§45), symmetry inside a definite integral (§51), floor and ceiling (§52), series and the method of differences (§62), recursions (§63), and counting and selections (§65). Curvature and the trapezium rule (§55) and coordinate distances and reflections (§58) were also absent before.

— **Paper 2 grew from three pages to a part of its own,** which is closer to the half of the exam it actually is, and gained the "exactly one of these is true" archetype (§72) and a systematic treatment of nested counterexamples (§69.2).

— **The error taxonomy in Book One §15 was rebuilt** from nine categories to thirteen, grouped into four families, with a remedy tree.

— **The whole book is now navigable.** Every contents row, every part-divider listing, every §NN cross-reference in the text and every running head is a live link in the PDF. Tapping the header at the top of any page returns you to the contents.

— **The community papers are named.** §27.1 lists the mock series worth sitting, and §27 names the worksheets, notes and adjacent banks rather than describing them generically. Where to find all of it is §80.

— **The dates box was withdrawn.** Deadlines, test windows and access arrangements are published by UAT-UK, they change every cycle, and a reader who trusts a stale deadline for access arrangements has been actively harmed by a document that was trying to help. What remains of that section is the part that was actually about preparation.

79.2 Corrections carried forward

The previous edition shipped with an errata list of ten items, which is a good habit and rarer than it should be. Those ten are carried through and are reproduced here in compressed form, followed by what this rebuild found on top of them.

1. The two-lobe cubic formula was mistyped, with a stray factor. The clean version is in §54.2.
2. A drill labelled as the quartic drill contained no quartic questions; every question in it used the cubic constant $|a|D^4/12$. The quartic results had no drill at all.
3. The $|a|D^5/20$ result never said which configuration it belonged to. It is the case $a(x-\alpha)^3(x-\beta)$, a triple root and a simple root, and §54.1 produces it in one line.
4. A statement about cubic end behaviour was filed under quadratics, in both the manuscript and the summary sheet.
5. The summary sheet attributed $S = |a|D^3/6$ to "any cubic and any secant". That constant belongs to a quadratic cut by a line; a cubic cut at three points has two unequal lobes.
6. The summary sheet mis-sourced the 15° triangle to "$\sin 45 \pm \sin 30$". Sine is not additive: $\sin 45^\circ - \sin 30^\circ \approx 0.207$ while $\sin 15^\circ \approx 0.259$. The correct source is $\sin(45^\circ - 30^\circ)$ expanded.
7. The general lobe formula on the summary sheet had a sign problem, writing the second lobe with a factor that is negative for $\alpha < \beta < \gamma$.
8. The repeated-root rule was stated the wrong way round. Even multiplicity keeps the sign, so the curve bounces; odd multiplicity changes it, so the curve crosses, flattening as the multiplicity grows (§43.3).
9. Two statements about end behaviour were written as products of infinities and compared with zero, which is not a well-defined operation. What is meant is a statement about the two **signs**, and that is how §43.1 now writes it.
10. The index-hopping justification did not match its example, giving $2a_5 = a_2 + a_8$ and justifying it with $a_7 - a_4 = a_4 - a_1$. Both statements are true; the check to use is that the indices balance, $2n = (n-k) + (n+k)$ (§61.2).

THREE FURTHER CORRECTIONS FOUND IN THIS REBUILD

A. An arithmetic slip in the arithmetic-progression composite. With $p = -\frac{1}{2}$ and $d = \frac{3}{8}$, the previous edition gave $S_{10} = 10\left(-\frac{1}{2}\right) + 45\left(\frac{3}{8}\right) = \frac{55}{8}$. The expression is right and the arithmetic is not: $-5 + \frac{135}{8} = \frac{95}{8}$. Corrected in §61.

B. A sign error in the tangent-circle composite. For a circle through $(0, 0)$ and $(6, 0)$ tangent to $y = x$, the previous edition reached "$t^2 - 6t + 9 = 0$, so $t = 3$". Squaring $|3 - t|/\sqrt{2} = \sqrt{9 + t^2}$ gives $9 - 6t + t^2 = 18 + 2t^2$, that is $t^2 + 6t + 9 = 0$, so $t = -3$. The **count** of circles was right; the centre was not. The centre is $(3, -3)$ and the radius is $3\sqrt{2}$, which you can verify directly: the distance from $(3, -3)$ to $y = x$ is $6/\sqrt{2} = 3\sqrt{2}$. Corrected in §76.

C. Outdated format advice in the absorbed primer material. The ten-lesson primer this edition draws on was written before the test moved on screen in 2024. Its mathematics is sound and its explanations are careful; its assumptions about annotating a printed paper, about timing, and about change of base being freely available are not. Everything taken from it has been re-tiered against the current specification and rewritten for a screen-based sitting. It is still worth reading in full for any topic that stays shaky, and it is listed in §27 as item 6.

IF YOU FIND ANYTHING ELSE

Send it in. It gets fixed and you get the revised file. That was the first printing's policy and it remains this one's, and the three corrections above exist because somebody did exactly that.

80 Where to find everything

Every link in this section is live in the PDF: tap it. Between them they cover the interface, the community papers, the drives that hold nearly everything else, and the place to ask when something is not working.

80.1 The five you will actually use

Sit every timed paper here tmua.io/#papers	A realistic Pearson-style interface for **every community paper**, with a timer, a flag button and a review screen. All your timed work from Stage 4 onwards belongs here, because sitting on printed paper trains a skill the exam does not test (§23).
TMUA9. tmua9.com	TMUA9.'s own site, co-founded by Viluxent. Mocks, worked analyses, and the source of the extraction framework that Part III is built on.
Resource drive, part 1 open drive one	Compiled by *qidalia*. Between them, the two drives collect nearly everything named in §27: official archives, community series, adjacent banks and topic notes.
Resource drive, part 2 open drive two	Start in the drives rather than searching. Hunting for material is not preparation, and the search cost is real (§28.1).
Ask when you are stuck discord.gg/AWTvwjDpsn	The TMUA9. Discord. Where new papers surface first and where a solution that makes no sense can be explained. Use it to unblock one question, never as a substitute for the hour you were going to spend extracting one.

80.2 The two that are authoritative

Everything above is community material and it is excellent. Two sources are not community material, and where they disagree with anything in this book, they win.

— **UAT-UK** publish the specification, the past-paper archive, the specimen and practice material, and the notes on logic and proof. They are the only authoritative source for what is examinable and for every date and deadline. Read the specification yourself, once, for an hour (§41).

— **Pearson VUE** deliver the test and publish the official practice and specimen tests in the real interface. Two hours there is the best return per hour in Book One (§23).

> **WHAT IS ACTUALLY YOURS**
>
> None of it is on that list.
>
> The thing that gets you the 9.0 is the notebook you fill with your own wrong routes, your own split points, your own critical thoughts written out word for word, and your own error mix tallied every Sunday. No document can hand you that, including this one.
>
> The purpose of a book like this is to stop you wasting months working out what to do. What to do is now settled. The doing is not, and it never was.

One page to keep

The content edition

AREAS, IN ONE IDENTITY

If the difference between curve and line is $a(x-\alpha)^m(x-\beta)^n$, the area is $|a|D^{m+n+1}\dfrac{m!\,n!}{(m+n+1)!}$ with D the gap in x. Constants: $(1,1)\to\frac{1}{6}$, $(2,1)\to\frac{1}{12}$, $(3,1)\to\frac{1}{20}$, $(2,2)\to\frac{1}{30}$.

Three intersections: split the far factor as $(x-\beta)-\Delta$. The tilt of the line never matters, because subtracting a line cannot change the leading coefficient.

THE THREE QUESTIONS THE DISCRIMINANT ANSWERS

How many solutions, does it touch, and is it always positive. $\Delta>0$, $=0$, <0. Always positive means $a>0$ **and** $\Delta<0$, and always check $a=0$ separately when a holds a parameter.

Sum of roots $-b/a$, product c/a, and every symmetric function comes from those two.

SUBSTITUTIONS CARRY CONSTRAINTS

$t=2^x$ needs $t>0$. $s=\sin x$ needs $|s|\le 1$, and then each s gives several x. $u=\sqrt{x}$ needs $u\ge 0$. $v=x+1/x$ needs $|v|\ge 2$. Every logarithm needs a positive argument. **The rejected root is on the option list.** Check the constraint, do not recite it.

INTEGERS

Squares are 0 or 1 mod 4, and 0 or 1 mod 3. $n(n+1)$ is even; $n(n+1)(n+2)$ is divisible by 6. Divisor count is $\prod(e_i+1)$; a square needs every exponent even. HCF takes the smaller exponent, LCM the larger. Count by naming disjoint cases, then the product rule, then check both boundaries and the empty case. Integers in $[a,b]$: $\lfloor b\rfloor-\lceil a\rceil+1$.

GRAPHS AND SYMMETRY

Mirror at $x=p$ if $f(p+t)=f(p-t)$; centre at (p,q) if $f(p+t)+f(p-t)=2q$. Average the two arguments to find p. Modulus inside is symmetry, modulus outside folds upward. Factor before interpreting $f(ax+b)$.

Over a symmetric interval, odd integrates to zero and even doubles. $\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$.

COUNTING SOLUTIONS

Sketch two curves and count crossings. Do not solve. Transform the **interval**, not the trigonometric equation. Slide the horizontal line past every stationary **value**.

PAPER 2, THE WHOLE MACHINE

If P then Q; Q if P; P only if Q; P sufficient for Q; Q necessary for P; not Q implies not P — all one statement, all $P\subseteq Q$. The stricter condition implies the looser one. Contrapositive always holds, converse never does for free. Negate mechanically, keeping the order; the negation of $P\Rightarrow Q$ is P **and** not Q. Mixed quantifiers do not commute. A counterexample needs p true and q false. A false hypothesis makes an implication true. “Exactly one is true”: normalise all of them, delete duplicates, delete anything that implies another.

BEFORE YOU COMPUTE ANYTHING

What can the answer **not** depend on? Scaling fixes the shape of the answer, translation kills anything mentioning position, parity kills half the integers, and sign and size usually kill two options in ten seconds. If you cannot confirm a shortcut’s conditions in five seconds, use the default and be right slowly.

CLOSING

Learn the trigger, price the move

Book Two is longer than Book One and it is worth less.

That is not false modesty. It is the whole argument of the first half, restated at the end of the second. Content is necessary and it is not sufficient, and a candidate who knows every result in here and cannot sit in front of an unfamiliar problem for four minutes without panicking will not score a 9.0. The reverse candidate might.

What content buys you is time. Ten questions under ninety seconds each is not a target about speed, it is a target about **room** — room for the four questions on the paper that nobody can prepare you for, the ones where you have to invent something in a test centre with a timer running. Every card in Book Two exists to buy back thirty seconds for those four questions.

So use this half as an index, not as a syllabus. Read a result, price it, and either put it in your deck or leave it on the page. A deck of ninety cards you actually review beats a deck of two hundred you abandon, and a result you half-remember is worse than one you never met, because the half-remembered one fires when it should not.

And when a question arrives that is in none of these sections — which will happen, three or four times, on the day — close this book in your head and go back to the other one. Name the object. Find the unused condition. Produce one example, then a second. Something will change.

See you all with those 9.0s.

Viluxent (mkb6) · TMUA9. (jamiebaek.)

Format and content checked against the current UAT-UK specification. Verify every date yourself.